

AN ARCHIVAL ALMA SEARCH FOR UNRESOLVED SPECTRAL-CARRIER TECHNOSIGNATURES TOWARD 60 STELLAR SYSTEMS WITHIN 40 PC AT 114–873 GHz

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ABSTRACT

We ask whether any nearby star operates a narrowband millimetre or submillimetre transmitter bright enough for ALMA to have recorded it incidentally. Technosignature searches have barely touched 90–873 GHz; published work lies almost entirely below 30 GHz.

We searched the ALMA archive for an unresolved, drifting carrier at a stellar position. The sample is 1651 windows toward 82 systems within 40 pc, all observed for other science, mostly debris discs. Of these, 402 windows toward 60 systems have channels fine enough to follow a drifting carrier. These cover 47.7 GHz of sky frequency, in disconnected intervals, not one continuous band.

Sensitivity was measured, not modelled, by injecting artificial carriers into the calibrated visibilities and recovering them through the same pipeline. The median system’s best window reaches EIRP₉₀ = 2.9×10^{15} W. A 12 m dish radiating 1 MW at 230 GHz appears at 5.9×10^{14} W, reached for 11 of the 60 systems.

The search returns 56 threshold crossings, and the unexplained ones are as numerous as chance alone predicts. Carbon monoxide accounts for 14, including the β Pictoris disc, recovered with no prior information. Only 2 crossings outrank all 512 spatial controls, and neither recurs at a second epoch. Stacking epochs for 69 stars in the stellar rest frame reaches 4.50×10^{12} W for Proxima Centauri, for carriers whose reflex drift stays within 0.635 km s^{-1} .

We find no persistent narrowband transmitter. ALMA chose these targets, so the result constrains these 82 systems, not the solar neighbourhood.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

In plain terms, this paper asks whether any nearby star is transmitting a narrow radio signal that ALMA happened to record while observing something else. ALMA’s public archive holds thousands of hours of millimetre and submillimetre observations taken for quite different reasons, mostly studies of dust discs and planet formation. Every one of them also contains a spectrum of the star at the centre of the field. A transmitter would appear in such a spectrum as a spike confined to one or two channels. It would sit at the star’s position and nowhere else in the field, and reappear if the star were observed again. We searched the archive for exactly that, with one pipeline and no human intervention. Nothing of the kind survived.

A technological civilisation may imprint the electromagnetic spectrum detectably whether or not it intends to, a *technosignature* (Sheikh 2020; Vidal et al. 2026). Millimetre wavelengths are an attractive place to look for one. Interstellar scattering and dispersion fall steeply with frequency, so a high-frequency carrier arrives less distorted. A given aperture delivers higher gain, and so more radiated power per watt supplied to the transmitter. Coherent millimetre emission is also already engineered on Earth, for directional links, radar and beamed

power.

The band is correspondingly hard to search. The atmosphere transmits unevenly and adds phase noise, and receivers are noisier than at centimetre wavelengths. Archived correlator channels are thousands of times coarser than a dedicated narrowband backend, so a carrier is diluted within one channel and its drift can smear it across several. These bands are also crowded with molecular lines, a dense background of real astrophysical signals that a threshold search must separate from an artificial one.

Six decades of radio searches have examined many thousands of stars, yet sample only 6×10^{-18} of the space of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018). Essentially all of that effort lies below ~ 30 GHz; above it we are aware of one published search, in ALMA Band 3 (Mason et al. 2025). Few searches at any frequency reprocess existing interferometric archives, although those archives already hold a spectrum of every target they have pointed at (§2).

This paper is an archival ALMA search of nearby stellar systems across that band, with its sensitivity and its selection measured rather than assumed. The signal class is an *unresolved spectral carrier*: emission confined to a few channels, unresolved at the stellar position, and persistent enough to recur between epochs. Each qualifier is a real restriction. A carrier broader than the channel width would not be recovered here, nor one that fills the synthesised beam, nor one that appears once and never again. The fine channels used are 15.3–1953 kHz wide, so “narrowband” here describes the assumed transmitter

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and never the resolving power of the data. Channelisation also divides the archive. Only 402 of the 1651 searched windows support the fine-channel search; the rest register an unexplained spike but cannot tell a drifting signal from a stationary one.

Every window follows the same four steps: a threshold crossing; attribution to a known line; comparison of the star with 512 control positions in the same field; and a search for recurrence at another epoch. Figure 2 is the roadmap. One worked case runs through the paper: a crossing toward β Pictoris, the largest star-versus-control contrast here, attributed to the disc’s known circumstellar CO in two transitions and 9 execution blocks (§5.3). Where this paper says “sensitivity” it means EIRP_{90} , the equivalent isotropic radiated power that the complete automated selection recovers nine times in ten. The internal trigger level P_{trig} is not a sensitivity.

Three limits on interpretation hold throughout. The targets were chosen by archival disc and planet-formation programmes, not for this search. The sample is therefore enriched in F and G stars and depleted in M dwarfs: only 21 of 5908 catalogued M dwarfs within 40 pc appear in it. Every statistical statement here concerns the ALMA archive-selected stellar sample, not the solar neighbourhood. The survey cannot constrain the local habitable-zone planet population, which M dwarfs dominate (§3.1). The search statistic and the line mask were defined on these data, so the rates quoted here are post hoc, and §4 sets out what was fixed when. And only 42 of 82 systems were observed more than a day apart, so a non-detection constrains a persistent carrier far better than an intermittent one.

2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first targeted experiments (Cocconi & Morrison 1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) all worked below a few gigahertz. The modern generation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay et al. 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026). The classical threshold search used here lacks that capability, and instead uses a spatial test on the star against control positions in the same field, which a single dish cannot make.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA observes in Bands 3 to 10, spanning roughly 84 to 950 GHz; of those eight bands, one has been searched for technosignatures. For continuity with the centimetre literature we note one familiar power scale: the Arecibo planetary radar radiated 2.0×10^{13} W at 2.38 GHz (Drake 1974; Sheikh et al. 2025). It carries no transmitter model into the millimetre band, where an aperture of that diameter is not realisable, so the benchmark used in §6 is instead

a dish of 12 m.

Mason et al. (2025) is the only prior millimetre search and the only comparable archival one. They searched Band 3 windows toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a 5σ threshold corresponds to $\text{EIRP}_{\text{min}} > 6.9 \times 10^{17}$ W. Their targets are four orders of magnitude more distant than ours, so an EIRP ratio between the two surveys would be almost entirely a distance term. The quantity that does not depend on distance is the minimum detectable *received* flux, $F_{\text{min}} = S_{\text{min}} \delta\nu$. It must be quoted at a matched channel width, because S_{min} falls as $\delta\nu^{-1/2}$ while F_{min} rises as $\delta\nu^{1/2}$. Mason et al. reach 3.8×10^{-23} – 5.8×10^{-23} W m $^{-2}$ across their four fields, on a 30.52 kHz channel at a bare 5σ threshold. On that same channel width and the same threshold convention, the fine-channel windows searched here reach 5.2×10^{-24} W m $^{-2}$ at best, a factor 9 deeper, and 7.1×10^{-23} W m $^{-2}$ at the median, a factor 1.5 shallower. The two experiments are therefore comparable in depth and differ in what they cover: distant stars in four calibrator fields, against 89 stars within 40 pc over a frequency range reaching about nine times higher. They also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the dense forest of molecular lines raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive data with the pipeline of §4, so for any star in common the numbers to cite are this paper’s.

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. The vertical axis is physically commensurable: for a carrier narrower than every programme’s channel, the minimum detectable EIRP is a total radiated power and not a spectral power density, so the same quantity is plotted for each programme. What differs is the convention. The value plotted for this survey is EIRP_{90} , a 90 per cent recovery through the complete automated selection, whereas most published limits are nominal thresholds at which recovery is not stated; the comparison runs against us. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one. Panel (b) shows the other axis that matters. Archived correlator channels here span about seven orders of magnitude in width, and all of them are far coarser than a dedicated narrow-band backend. The distinctive feature of this survey is therefore its frequency regime and its channelisation, not its depth. It also samples far fewer stars and far fewer epochs than a dedicated search, and complements those programmes rather than deepening them.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth’s own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

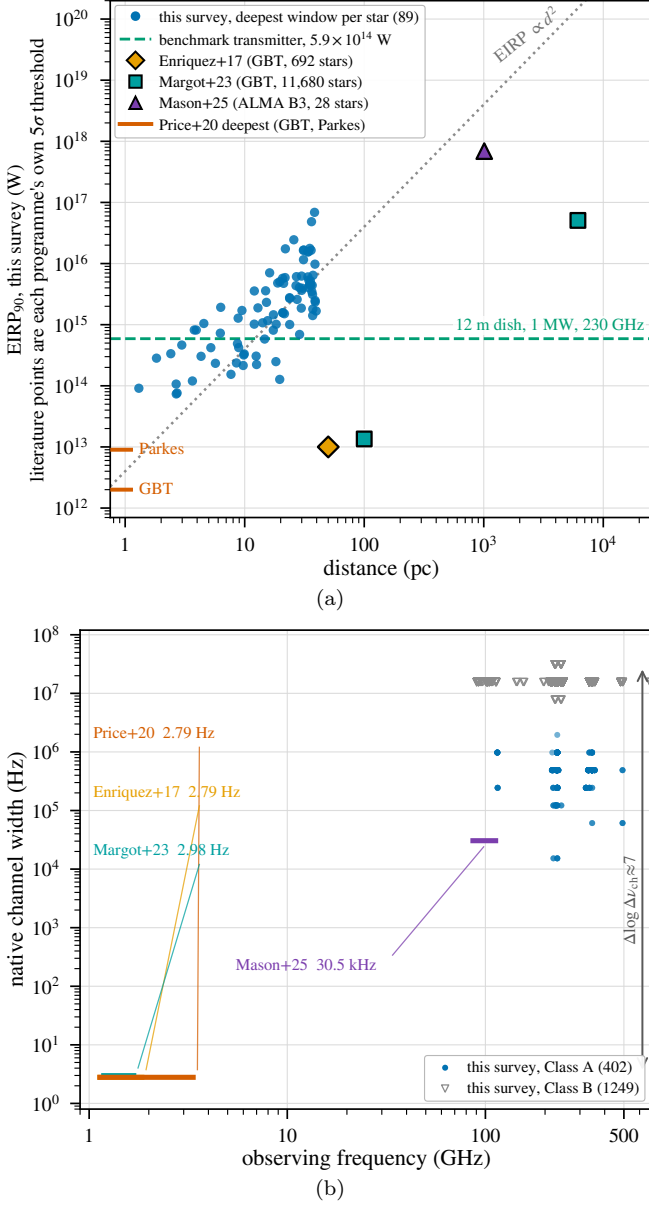


FIG. 1.— Where this survey sits among published radio technosignature searches. (a) Minimum detectable EIRP against distance: blue, this survey, the deepest window per star; orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020). Grey dotted, $EIRP \propto d^2$ at constant received flux; green dashed, the 12 m benchmark transmitter of §6. The ordinates are commensurable as total powers for a sub-channel carrier; what differs between programmes is the detection threshold and the completeness level each adopts, and this survey’s value is the stricter of the two conventions. (b) Native channel width against observing frequency for every searched window (filled, Class A; open, Class B), with the comparison programmes as horizontal segments. The channel widths span about seven orders of magnitude, which is what the $\Delta \log \Delta \nu_{ch} \approx 7$ annotation marks: the channelisation, not the plotted power, is what differs most between programmes.

3. SAMPLE SELECTION

We searched 1651 spectral windows toward 89 stars in 82 independent systems within 40 pc, drawn from 404 public ALMA execution blocks and 4.1 TB of raw data through one frozen pipeline. The sample is two experiments, not one. 402 windows toward 60 systems are fine-channel, called *Class A* throughout: these resolve a

drifting carrier and are the primary experiment, and every sensitivity and every exclusion quoted in this paper refers to them unless stated otherwise. The remaining 1249 windows are coarse-channel, *Class B*, and support a secondary search for unresolved spectral excess; they can register a spike but cannot distinguish a drifting signal from a stationary one. Table 2 collects the survey as searched.

The inclusion criteria are objective and were applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent. It must also have its proper-motion-corrected position at the observation epoch inside the quoted field of view of at least one public ALMA execution block. Distances are direct inversions, $d = 1/\varpi$. The star need not sit at the pointing centre. Disc, planet and spectral type enter as descriptive metadata only, and no star or window was excluded by an unlisted criterion or for short integration. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 89 yielded a searchable window (Fig. 2). The complete per-star and per-window ledger, including the five-criterion decision tree and every edge case, is in the data release.

3.1. The archival selection

This sample was not designed. It is whatever previous ALMA programmes happened to observe, for their own reasons, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. The consequence is a strong and measurable bias, tabulated against a volume-limited reference census in Table ??: the sample is enriched in F and G stars and depleted in M dwarfs. Only 21 of the 5908 catalogued M dwarfs within 40 pc appear in it at all, and only 13 have Class A coverage, the only coverage that can resolve a drifting carrier. Coverage is also concentrated: 285 of the 1651 windows come from 3 systems. Every statistical statement in this paper is therefore a statement about an archive-selected sample of nearby stars. None of them is an occurrence rate for the solar neighbourhood, and because M dwarfs host most of the nearby habitable-zone planets, this survey cannot constrain transmitters around the kind of star most likely to have one.

3.2. Frequency coverage and archival scope

The Class A experiment covers $\Delta \nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: it is 26 disjoint islands scattered between 114.0 and 873.1 GHz, not a band. The endpoints are a range over which observations exist, never a range that was covered. Adding the coarse-channel windows brings the total archival coverage to 118.1 GHz in 46 intervals, of which 113.3 GHz survives the molecular-line mask of §4. The distribution is shown in Fig. ??; a transmitter at a frequency between the islands was never observable here, whatever its power.

Of the 177 execution blocks in scope for this search, 144 were calibrated and searched. The shortfall is accounted for rather than estimated. For 18 blocks, 237 GB in all, the archive holds only raw visibilities that were never pipeline-calibrated; this is the single largest gap, and no effort on our side would close it. A further 11 blocks

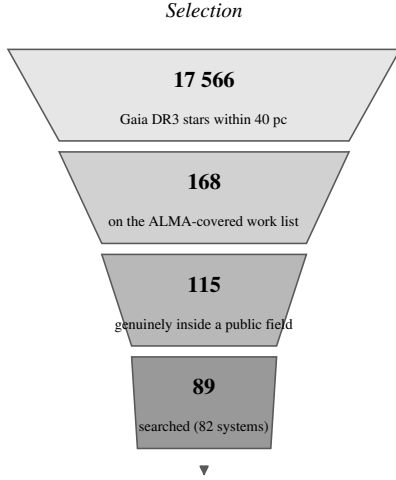


FIG. 2.— The selection funnel, from the catalogued population within 40 pc to the stars with a searched window. The three quantities the text keeps apart are the 168-entry work list the archive query returns, the 115 stars a public field genuinely contains, and the 89 that yielded a searchable window. The chain applied to what comes out of the funnel is Fig. 5.

TABLE 1

SYSTEMS OF PARTICULAR INTEREST: EVERY SAMPLE SYSTEM WITHIN 15 pc HOSTING AT LEAST ONE CONFIRMED PLANET, ORDERED BY DISTANCE. N_{pl} IS THE CONFIRMED-PLANET COUNT, N_A THE NUMBER OF CLASS A WINDOWS, $EIRP_{90}$ THE DEEPEST CLASS A SENSITIVITY REACHED, AND N_{eb} THE NUMBER OF EXECUTION BLOCKS COVERING THE SYSTEM, SO $N_{eb} > 1$ MEANS REPEAT EPOCHS EXIST. SIX OF THESE SYSTEMS HAVE NO CLASS A COVERAGE AND SO CARRY NO DRIFT-RESOLVING LIMIT — THE ARCHIVAL SELECTION OF §3.1 ACTING ON EXACTLY THE TARGETS A TRANSMITTER SEARCH WOULD CHOOSE. THE FULL TARGET LIST IS IN THE DATA RELEASE.

System	d (pc)	N_{pl}	Bands	N_A	$EIRP_{90}$ (W)	N_{eb}
Proxima Cen	1.3	2	6	0	—	25
Barnard’s Star	1.8	4	6	0	—	3
τ Cet	3.7	3	6	6	2.3×10^{14}	8
BD+05 1668 (GJ 273)	3.8	2	6	0	—	25
HD 33793	3.9	1	6	0	—	22
GJ 674	4.6	1	6	0	—	20
HN Lib	6.3	1	6	5	7.3×10^{14}	5
GJ 581	6.3	3	6	0	—	9
61 Vir	8.5	3	7	5	2.4×10^{14}	5
GJ 849	8.8	2	6	6	1.3×10^{15}	6
HD 285968	9.5	1	6	4	1.7×10^{15}	4
AU Mic	9.7	4	3,6,9	13	2.1×10^{14}	14
TRAPPIST–1	12.5	7	3,6,7	5	3.1×10^{14}	18
HD 69830	12.6	3	6,7	22	2.2×10^{14}	8
LHS 1140	15.0	2	6	4	3.6×10^{15}	4

failed calibration, 3 were not attempted, and 1 needed more working disk than was available.

4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star’s own position. Such a carrier is narrow compared with a single correlator channel. Its sky frequency drifts, because the transmitter accelerates along the line of sight. Over a long track its power is therefore spread across many channels, and a search that treats each channel on its own will miss it. The analysis de-drifts the data before it measures them.

4.1. The de-drifted search statistic

TABLE 2

THE SURVEY AS SEARCHED. EVERY FREQUENCY UNION QUOTED HERE IS AN AGGREGATE OVER CLUSTERED, REPEATEDLY REVISITED COVERAGE, WHOSE DISTRIBUTION IS FIG. ??; NO UNION SHOULD BE READ AS CONTINUOUS COVERAGE BETWEEN ITS ENDPOINTS. POWERS AND FREQUENCIES ARE GIVEN TO TWO SIGNIFICANT FIGURES, EXCEPT WHERE A CHANNEL OR SKY FREQUENCY MUST BE MATCHED TO THE RELEASED CATALOGUE AND IS GIVEN TO THE PRECISION THE GRID SUPPORTS.

Quantity	Value
Stars searched; independent systems	89; 82
work-list entries; genuinely covered	168; 115
Execution blocks; distance range	404; 1.30–39.63 pc
Spectral windows searched	1651
Class A fine / Class B coarse	402 / 1249 (76% Class B)
12 m array; ACA 7 m	597; 1054
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
$\Delta\nu_A$, unique Class A frequency union (the drifting-carrier experiment)	47.7 GHz in 26 islands, 114.0–873.1 GHz
coverage Class B adds	70.4 GHz
total archival coverage	118.1 GHz (46 intervals)
usable after the molecular-line mask	113.3 GHz
Nominal trigger P_{trig} , Class A;	2.2×10^{13} – 6.9×10^{16} W;
Class B	1.6×10^{13} – 2.8×10^{17} W
Execution blocks in scope; searched	177; 144

Each window’s dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\nu, \dot{\nu}} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν , and $\sigma(t, \nu)$ is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier. Its numerator adds the signal coherently along the track and its denominator is the noise of the stacked spectrum, so T is an absolute signal-to-noise ratio and not a local contrast. The noise scale is the median absolute deviation across 512 control positions, taken at each integration and each channel. It is therefore local in time and frequency, global in position, and shared by the star and every control. The across-position median is never subtracted from the numerator. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. T_* denotes its value at the star.

Spectra are extracted at the star’s apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position: an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA’s own calibration and quality assurance, with their flagging carried through and with no regridding and no deconvolution. All delivered channels are searched, with no edge-channel trimming. A per-integration median along the frequency axis removes the continuum; no median is taken along time, so a one- or two-channel feature cannot be removed by it. One subtlety matters for every power quoted below. The de-

livered channels are smoothed, so a channel’s effective noise bandwidth is not its separation, and flux density times channel width is not yet a radiated power. A response correction computed from the correlator’s own channel shape supplies the difference. It is derived, and validated against the effective resolutions ALMA reports, in Appendix A.

Two power quantities carry the main text. The trigger level P_{trig} is the equivalent isotropic radiated power, $\text{EIRP} = 4\pi d^2 S \Delta\nu$ at distance d , of a carrier that makes $T_\star$ reach $T_{\text{trig}} = 5$ in one native channel of width $\Delta\nu$. It is an internal pipeline level, the power at which the search fires, and it is neither a sensitivity nor a significance. The sensitivity is EIRP_{90} , the power the complete automated selection recovers nine times in ten, and it is measured rather than assumed. 8860 synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline. They span 52 block-window configurations, both ALMA arrays, four bands and the sample’s full range of channel widths. For Class A the result is $\text{EIRP}_{90} = 5.70 P_{\text{trig}}$. Every limit quoted in this paper is an EIRP_{90} , given to two significant figures beside its native channel width; its error budget is in Appendix ???. The absolute flux scale has been checked twice from outside. Our extractor reproduces an independent reduction of the same AU Mic visibilities to 0.994 ± 0.014 on a 29.4 mJy point source. It reproduces the published quiescent flux density of ϵ Eridani to 1.10 ± 0.31 (Burton et al. 2022). The two checks share neither band nor array.

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units, $a_{\text{los}} = c \dot{\nu}/\nu$, so a ceiling expressed as $\dot{\nu}/\nu$ is the same physical limit in every band. The default ceiling is $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$, that is $|a_{\text{los}}| \leq 3.60\text{--}4.00 \text{ m s}^{-2}$. That is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more. The margin is not cosmetic: at unit margin the search would have been blind to two of the windows that produced a crossing. The grid is uniform in $\dot{\nu}$, and fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track. Two limits are worth stating plainly. The rates searched are the apparent ones at the telescope, so a deliberately chirped carrier is not distinguished from an accelerating platform. And a transmitter need not be bound to a known planet at all, so drifts faster than the ceiling are simply unconstrained (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division. Channelisation in the sample is bimodal, with an empty gap between the two modes, so the split is unambiguous. Class A is the primary experiment: the 402 fine-channel windows, median 488.3 kHz. In these the maximum searched drift crosses of order one channel or more over the on-source track, the condition $\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta\nu_{\text{ch}} \gtrsim 1$. Class A is also the only class with measured end-to-end recovery of a drifting carrier. Class B is the secondary experiment: the 1249 coarse-channel windows, mode 15.625 MHz, in which a carrier

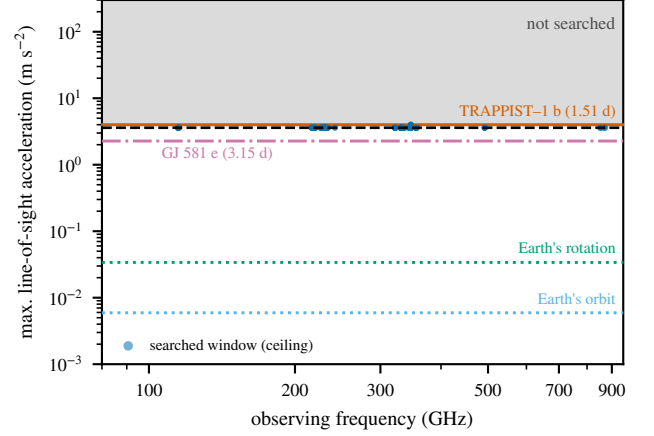


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 402 Class A windows (points), converted to line-of-sight acceleration, $|a| = c \dot{\nu}/\nu$, against observing frequency; the shaded region above is unsearched. The boundary is flat because the grid was built at constant acceleration; in the units the search works in it is not, running 1.4 kHz s^{-1} at 115 GHz to 10.5 kHz s^{-1} at 872 GHz. Horizontal lines are reference accelerations: Earth’s rotation (0.034 m s^{-2} at the equator) and orbit (0.0059 m s^{-2}), and the two most demanding known planets of any target, TRAPPIST-1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}). The boundary drawn is the default ceiling, 3.60 m s^{-2} , which is also the median; it runs to 4.00 m s^{-2} on the 5 TRAPPIST-1 windows, and it is each window’s own ceiling that decides whether a given orbit is contained. The most demanding case is TRAPPIST-1 b, whose line-of-sight acceleration exceeds the default ceiling; TRAPPIST-1 transits, so the edge-on figure is the relevant one, and planet b lies above the default ceiling for 29 per cent of its orbital phase and above the widened one for 1 per cent. It is searchable through the rest of the orbit, and planet c reaches the default ceiling only at modest eccentricity.

stays inside one channel for the whole track. These constrain persistent unresolved spectral excess, carry no drift discrimination, and are never described as a carrier search here. Each window’s own η_{drift} , trial-drift count and acceleration ceiling are released catalogue columns. A reader may therefore reclassify on the physical criterion rather than the instrumental one. The two criteria do not partition the survey identically, and we do not claim that they do. Results are given per class, and combined only where that is statistically appropriate.

4.2. From a crossing to a candidate

Every window goes through the same four tests, in this order. Only three words describe what comes out of them (Fig. 5).

- (i) *Crossing.* A channel $\times$ drift cell at the stellar position reaches $T_\star \geq T_{\text{trig}}$. This test generates the crossings; the other three dispose of them.
- (ii) *Line attribution.* The crossing’s frequency is compared with catalogued molecular transitions in the *star’s* frame. Within $\pm 50 \text{ km s}^{-1}$ of one of them the crossing is astrophysical and goes no further. The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, $\lesssim 13 \text{ km s}^{-1}$, with margin for the star’s systemic velocity. The mask classifies rather than detects: it is applied after the search, so no threshold moves and no window goes unsearched. Its cost is that the survey constrains nothing inside the excluded velocity intervals. Of the 118.1 GHz of unique sky frequency covered,

≈ 113.3 GHz survives the mask (Appendix C).

- (iii) *Spatial screen.* The stellar statistic is compared with the same statistic at 512 control positions. They lie in an annulus placed by each block’s own synthesised and primary beams, so that no control sits on the star. A crossing in which the star exceeds every control is a *candidate*.
- (iv) *Recurrence.* A candidate that repeats at the same tuning in an independent epoch of the same star is *confirmed*. This is the survey’s only confirmation criterion, and the archive’s uneven temporal sampling is the main reason it is a weak one.

Most crossings are expected by chance, because a fixed threshold is not a fixed false-alarm rate. The number of independent channel $\times$ drift cells in a window runs from 112 to 6.87×10^6 . The same 5σ level therefore buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. The expectation is computed window by window, from that window’s own cell count and the exceedance rate measured at its own control positions, then summed over the windows searched. It predicts 34.5 unattributed Class A crossings against 35 observed, about two-thirds of the 52 Class A crossings. T is not shown to follow any standard distribution under the null, which is why T_{trig} is a trigger level and never a tail probability. Appendix D gives the conditioning of the expectation and the control ensemble it is measured on.

The star-versus-control rank is an empirically useful candidate-prioritisation statistic, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection. The evidence for that is out of sample and independent of the crossings. On 315 windows the frozen pipeline had never seen, the stellar rank has median 0.405 where 0.5 is expected if the star and its controls were interchangeable. The displacement survives clustering by execution block. Its source is geometric: the control annulus samples a different radial and noise geometry from the star. A parameter-free correction, tested prospectively against that geometry, does not restore a validated null and is not adopted (Appendix B). The screen also loses power exactly where circumstellar gas is brightest, because a line filling the primary beam drives the star and the annulus up together rather than cancelling. Throughout this paper p denotes a genuine significance level, and no probability computed against the screen’s own null is called one.

Three corrections are applied before the statistic is formed. The extraction field is chosen so that the star lies inside the delivered primary-beam field, not merely inside the quoted field of view. The stellar position is propagated to the observation epoch and used as the apparent position. On-source time is accounted per integration from the delivered exposure records rather than from the scan intentions.

A visibility-domain check asks whether a candidate’s emission is at the stellar coordinates rather than elsewhere in the field. It is applied to every crossing and reported beside the chain rather than inside it, because it cannot discriminate at threshold: a noise peak at the star also has flux at the star. It is described in §5.5.1.

Two demonstrations establish that the chain finds what it is meant to find. Synthetic drifting carriers injected into real visibilities are recovered through the unmodified pipeline (Appendix ??). And β Pictoris CO, a real astrophysical feature at a known stellar position, is recovered and localised in two transitions and 9 execution blocks. The second is the survey’s positive control, and it is discussed with the results.

5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 402 fine-channel windows toward 60 systems. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1249 coarse windows. Together they are the 1651 windows of the released catalogue. **No carrier was found toward any of the 82 systems searched.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

5.1. Sensitivity: the power at which a carrier would have been found

The sensitivity of this survey is a measured completeness and not a threshold. Artificial carriers of known amplitude were added to the real calibrated ALMA visibilities, each with a random sub-channel phase and a random drift rate from its window’s own grid. They were then recovered through the unchanged pipeline: extraction, baseline removal, de-drifted stack, trigger, and the 512-position spatial screen. The power at which nine carriers in ten are recovered is EIRP_{90} . It is the only sensitivity quoted in this paper. It charges for the whole chain the analysis applies, because a carrier counts as recovered only when it reaches the trigger *and* the stellar position then outranks every control.

Measured that way, Class A reaches ninety per cent recovery at $\text{EIRP}_{90} = 5.70 P_{\text{trig}}$ and Class B at $4.55 P_{\text{trig}}$. Here P_{trig} is the window’s own 5σ trigger power. In absolute terms the median Class A window reaches 5.3×10^{15} W of equivalent isotropic radiated power. The median system, on its best window, reaches 2.9×10^{15} W, with a spread across systems of 1.3×10^{14} – 6.9×10^{16} W (Fig. 4a); that per-system median is the figure quoted in the abstract and in the conclusions. Statistical and systematic terms combine to one interval, -39 to $+21$ per cent, in which the largest systematic is the sub-channel phase term (Table ??). Two one-sided biases cannot be added to an error and are carried separately; together they are -5 to $+22$ per cent.

One caveat dominates the rest, and it is a limit of computation rather than of the data. Every injected window is re-extracted and re-searched end to end at each amplitude. Carriers were therefore injected directly into 13 of the 402 Class A windows, and the recovered ratio was transferred to the remaining 389. The spread of that ratio across the directly injected windows is measured rather than assumed: $\times 0.94$ – $\times 1.09$ of their median, carried in the budget as an uncertainty. Every value above should therefore be read as an upper bound on reach rather than a central estimate. Two further properties of the measurement bound how far it can be pushed. First, recovery does not reach unity: a bright source at the stellar position leaks through the dirty beam into

TABLE 3

THE LINE ATTRIBUTION AGAINST THE WIDTH OF THE MASK, EVALUATED IN THE STELLAR FRAME OVER ALL 56 CROSSINGS. THE ADOPTED HALF-WIDTH IS IN BOLD. THE LAST COLUMN COUNTS THE UNATTRIBUTED CROSSINGS THAT ALSO OUTRANK EVERY ONE OF THE 512 CONTROL POSITIONS IN THEIR OWN WINDOW. DOUBLING THE HALF-WIDTH CHANGES NOTHING; HALVING IT MOVES ONE CROSSING.

Half-width (km s ⁻¹)	attributed	unattributed	rank-flagged
±13	13	41	3
±20	14	40	2
±30	14	40	2
±50	14	40	2
±100	14	40	2

the control ring and raises the control maximum with it, from 4.8σ on an uninjected pass to 8.8σ at the brightest rung. Second, the ninety-per-cent point lies near the end of the injected ladder, so it is the power at which the measurement runs out rather than a sharp threshold.

5.2. The nearest stars: Barnard’s Star and Wolf 359

Barnard’s Star is the second-nearest stellar system, and Wolf 359 the third if brown dwarfs are excluded. To our knowledge these are the first millimetre or sub-millimetre technosignature limits for either; the searches checked are Enriquez et al. (2017), Price et al. (2020) and Margot et al. (2023). Both are non-detections, and only Band 6 data enter the limits. Quoted on the convention of §5.1, Barnard’s Star is constrained to 2.8×10^{14} – 4.3×10^{14} W over 12 windows. Wolf 359 is constrained to 3.4×10^{14} – 4.9×10^{14} W over 12.

5.3. Threshold crossings and how each was disposed of

The result is a chain of four tests, and it is read in one direction. Of the 1651 windows, 56 contain a channel $\times$ drift cell at the stellar position reaching $T \geq 5$. Such a cell is a *threshold crossing*. The stellar-frame molecular-line mask, ± 50 km s⁻¹ wide and fixed before any crossing was classified, identifies 14 of them as carbon monoxide. 40 are unattributed by that test. A further 2 await a per-block frame term and already fail the spatial screen, so the unattributed count has an upper bound of 42 and no conclusion turns on them. Of the unattributed, 2 outrank all 512 control positions in their own window. Neither of those two recurs in any later epoch covering its frequency. **No crossing survives the chain, and this survey reports no detection.** Figure 5 carries the chain with the number surviving each step. Table 4 is the crossing-by-crossing ledger and Table 5 the same result in counts.

The mask is informative rather than permissive, and its half-width is not a tuned parameter. The width follows from the Keplerian velocity bound on circumstellar gas in these systems, and the counts are flat around it. Doubling the half-width changes none of them; halving it moves one crossing from the attributed column to the unattributed one (Table 3). Evaluated in the stellar frame the mask attributes 14 crossings, where its occupancy of the searched band would give 2.3 by chance coincidence alone. The attributions are therefore a kinematic result and not an accident of bandwidth.

5.3.1. Most of the crossings are expected by chance

A 5σ trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel $\times$ drift cells

per window spans several decades, so a common threshold buys very unequal protection. Summing each window’s own measured false-alarm probability over the covered Class A windows gives 34.5 expected unattributed Class A crossings, against 35 observed. *About two thirds of the crossings this survey reports are expected by chance*, which is why the chain never treats a crossing as a detection and why no artefact hypothesis is needed to account for their number.

5.3.2. One execution block accounts for four of the crossings

4 crossings belong to BD+05 1668 (GJ 273, Luyten’s Star), with the highest statistics in the survey outside β Pictoris and HD 48370. All 4 lie in one ACA execution block, A002_Xc7fa6f_X28a8, in which the whole field is above threshold: 512 of 512 control positions exceed $T = 5$, against a survey median control maximum of 2.75. The single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy), but the residual does not integrate down; a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal. A slow drift common to every channel and every position therefore inflates the statistic across the field. The 24 sibling executions of the same field, with the same array and the same spectral setup, reach a maximum of $T_\star = 4.26$ and produce no crossing. *The median of the control ensemble, not its maximum, is the quality criterion this kind of search needs.* We report it as a diagnostic and do not adopt it as a cut, because a cut strict enough to remove this block would also remove the HD 48370 carbon-monoxide control.

5.3.3. Stellar flares are excluded by a broadband prediction

The hosts are not a quiet sample: 2 of 2 carry activity indicators, and magnetically active stars do produce coherent bursts at centimetre wavelengths. A millimetre flare, however, raises every channel of a window by the same flux density. A flare able to produce a 5σ single-channel crossing must therefore produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs 49–4,717 σ , while the observed continuum statistic has median -0.29 and $|Z| < 3$ in 45 of 48. One block settles it directly. AU Mic A002_Xa42f75_X319 contains both a crossing ($T_\star = 5.27$) and the published millimetre flare of MacGregor et al. (2020). In the crossing’s own time window the continuum is raised at $Z = 10.3$, while the crossing’s de-drifted track reaches only $Z_{\max} = 2.1$. *A real, bright, independently published millimetre flare in the same data contributes 0.17 σ to the single-channel statistic.*

5.4. β Pictoris: an astrophysical positive control, and the obstacle it identifies

A search that finds nothing must show that it could have found something, and the survey’s strongest features show exactly that. The frozen pipeline recovers the β Pictoris circumstellar carbon monoxide blind, without being told the disc is there. It recovers it in two rotational transitions and in 9 independent execution blocks spanning 8.4 yr: 5 Band 3 windows at $T_\star$ up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain, the Band 3 and Band 6

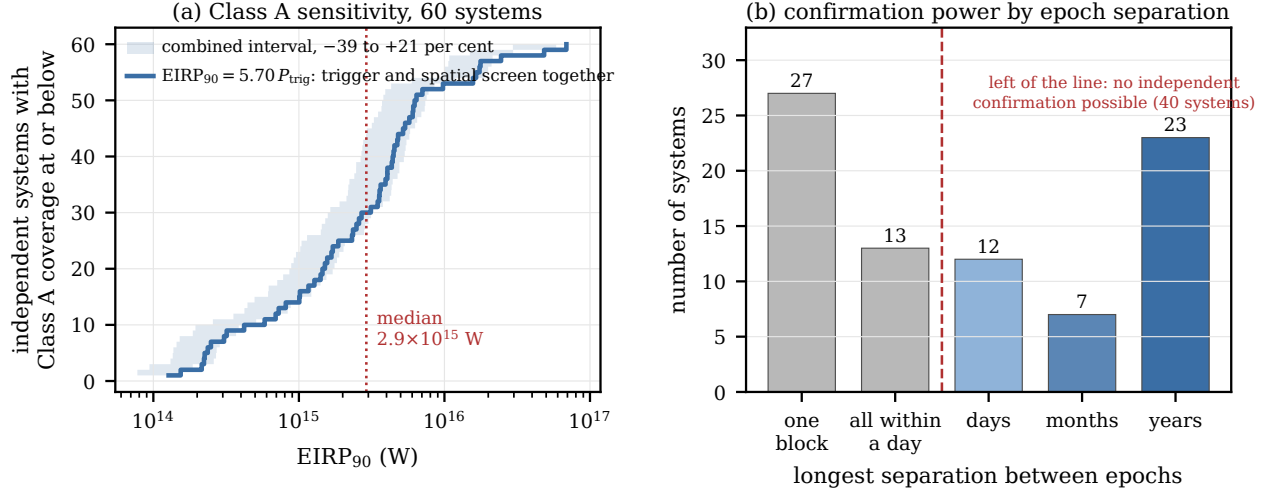


FIG. 4. — (a) Class A sensitivity: the cumulative number of systems whose best window reaches a given EIRP_{90} , the power at which nine carriers in ten are recovered through the trigger and the spatial screen together. The shaded band is the combined interval of Table ??.

(b) The temporal selection function: systems grouped by the longest separation between their searched epochs, an independent epoch being a separation of more than 1 day. **For the 27 systems observed in a single block, and the 13 whose blocks all fall within one day — left of the dashed line — independent confirmation was not possible at all, whatever the sensitivity in panel (a).** Recurrence over years can be tested for 23 systems of 82.

crossings lie -0.39 and -0.29 km s^{-1} from the stellar systemic velocity, within 0.39 km s^{-1} of it. They lie in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block’s own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

β Pictoris is also the control that shows the recurrence test of §5.6 working in the positive direction. Every β Pictoris carbon-monoxide feature the search flagged is recovered in an independent execution block, and the screened Band 6 window returns $T_{\star} = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_{\star} = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible foreground cloud emission. The block’s $^{13}\text{CO}(2\rightarrow1)$ window is bright at the ring and faint at the star. Extended line emission is therefore rejected by the same spatial statistic that promotes β Pictoris.

Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity. The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 14 of the 56 crossings are molecular lines. The mask that removes them costs almost nothing in bandwidth: 4.1 per cent of the unique-frequency union and 4.7 per cent of gross bandwidth, over 11 species. It is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth. Against a wide spectral-line harvest, rather than the 11-species mask used here, a *uniformly random* frequency in one of these windows falls within

$\pm 50 \text{ km s}^{-1}$ of some catalogued transition most of the time. A near coincidence is therefore the default outcome and carries no information either way. Any survey at these frequencies that quotes a line coincidence should quote the occupancy of the list it used.

5.5. The two crossings that pass the spatial screen

2 of the 56 crossings are both unattributed and out-rank every control in their own window, and both are in Band 7. 61 Vir, in block A002_Xc079b5_X82f, reaches $T_{\star} = 5.96$; CP-72 2713, in block A002_Xff0235_X4a6d, reaches $T_{\star} = 5.81$. Their frequencies and velocity offsets are in Table 4. Neither lies within $\pm 50 \text{ km s}^{-1}$ of a masked transition in its star’s own frame. *Each is disposed of by the same thing: its own later epochs.* 61 Vir has 7 further execution blocks covering the event’s frequency, and the largest statistic anywhere in them is $T_{\star} = 2.94$. CP-72 2713 has 5 such blocks. The deepest was taken two hours later and is 6.3 per cent deeper in noise; a carrier persisting at the discovery flux would have given $T_{\star} = 6.21$ there, and the measured value is 3.60. That excludes persistence at 6.8σ . Two further measurements close CP-72 2713 independently of any line list. The feature is wider at the recovered drift than the instrumental response, whereas a warm circumstellar sulphur line would be resolved. And CO(3→2) lies in the same window and is absent, below a limit fainter than the crossing itself.

5.5.1. The visibility fit, and why it is not the discriminant

Each crossing was also re-measured in the visibility domain, by phase rotating to the star and following the recovered drift. That result is reported in the data release beside the chain rather than inside it, because it is a consistency check on position and epoch. The reason is structural: a noise maximum *at* the stellar position has flux at the stellar position. At the trigger it therefore returns a real part close to the trigger itself, and carries almost nothing beyond the trigger that selected it. The injection campaign measures this directly. For

real sources at the star the two estimators track to within about ten per cent, and in the bin the unattributed crossings occupy injected sources return $\text{Re}/\sigma = 6.58 \pm 1.97$. The values returned by the two screened crossings are consistent with a real source and do not exclude one. **The discriminating evidence is the line mask, the spatial screen and recurrence, and not the visibility fit.**

5.6. Recurrence is the confirmation criterion, and the sample barely supports it

A persistent transmitter should be there the next time the same frequency is observed. That is the confirmation criterion used here, and it is applied in the star's own frame. The expected channel is computed by correcting the observed frequency for the Earth's motion at the repeat epoch and for the star's systemic velocity. The statistic is then read at that channel, at a single drift rate. Recurrence has been evaluated for 10 crossings over 51 covering repeat blocks. The largest statistic returned anywhere in them is $T_\star = 3.94$, against a trigger of 5. *Nothing recurs.*

Those repeat epochs include an archival extension searched with the same frozen pipeline, after the detection statistic and the line mask were fixed: 115 further execution blocks, 463 windows toward 16 stars in 16 systems. Every block in it is repeat coverage of a star and a tuning already in the sample, so it adds no star, no system and no frequency. What it adds is epochs. It is reported separately from the primary census, because every statistic above rests on a denominator and a crossing list fixed before these data were examined. Merging the two would destroy that property. Nor is the archive exhausted: 18 further in-scope blocks expose no pipeline calibration product in the ALMA delivery at all, so the recurrence statements rest on the epochs actually in hand (Appendix ??).

η Crv shows why the frame correction is not a detail. Its crossing at 357.281464 GHz is covered by exactly 2 epochs in the whole archive, and the second is 2.19 times deeper than the discovery. Matching in the stellar frame moves the expected cell by -80.2 channels, so a sky-frequency test would have examined the wrong channel. At the matched cell the statistic is -0.618, where a carrier persisting at the discovery flux would have given 10.95. *A persistent source is excluded at 11.57σ from one epoch*, which carries 4.79 equivalent discovery epochs of sensitivity.

The limitation belongs in the same breath as the criterion. Only part of the sample has repeat epochs separated well enough to test persistence at all. 27 of the 82 systems were observed in a single execution block, and 13 have all their blocks within one day. For none of those systems was confirmation possible, whatever the sensitivity (Fig. 4b). Only 23 systems can be tested over years, and for the two screened crossings the usable baselines are 2–769 d. **These tests exclude a carrier persistent over hours to days; they bound no duty cycle measured in months or years.** The constraint on an intermittent transmitter is accordingly much weaker than the constraint on a continuous one. Repeat coverage is also matched on the window rather than on the crossing frequency, which the release records only where

the pipeline stored a peak.

5.7. A stellar-frame multi-epoch stacked search

Combining the epochs of a star in that star's own rest frame deepens the limit. 69 of the surveyed stars hold more than one searched execution block. Over 368 stacked groups, 2319 epoch-windows and 1484 h, the flux limit improves by a median factor 1.46 over each star's best single epoch (2.39 at the 90th percentile, 4.45 at most). Nothing survives the criterion except known molecular lines. The limit a stack reaches is quoted here on the same completeness convention as every other sensitivity in this paper, measured by injection into the stack itself, and not as a threshold on a stacked spectrum.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs must be registered before they are added. The observer's motion and the star's systemic velocity move the observed frequency of a stellar-frame carrier by tens of km s^{-1} between epochs. The barycentric term alone swings by a median 8.2 km s^{-1} between the epochs of one star, and by as much as 53.9 km s^{-1} , which is many channels at the 488 kHz Class A channel width. Added as observed, the epochs of a real carrier would land in different channels and dilute rather than reinforce. Each epoch's frequency axis is therefore transformed to the stellar rest frame before stacking. The transform is applied per execution block rather than per star, because within a block the correction moves only 0.072 km s^{-1} and a single mid-time suffices. Epochs are then divided by their own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, with a minimum of 2 blocks and 64 common channels. The combination is thus a matched filter across epochs rather than a rescaling of one of them. Groups are keyed on the star, resolved externally, and not on the archival directory the search ran in: 78 directories resolve to 69 stars and 394 groups to 368, with no star lost and none gained. Merging identities does not merge windows. A group is still the intersection of the windows covering one stellar-frame frequency, so two tunings sharing no channel cannot combine. 27 of the 2838 retained products (1.0 per cent) carry weights up to 10^{11} times the median and are excluded.

The criterion was fixed before any stellar position was examined. A group is a candidate if its stacked $Z \geq 6.0$ at a common channel and above every stacked control maximum. Two further clauses guard against one epoch carrying the result: no single epoch may supply more than 0.60 of the weighted sum, and at least 0.50 of the signal must survive removal of the strongest epoch. Controls are stacked identically and every clause is evaluated at them as well. That measures the false-alarm rate of the whole test at 0.21 per cent, against the 1 per cent ceiling fixed in advance.

Registration is verified on real lines before it is used on anything else. The per-block transform collapses HD 285968's molecular line from a 19.26-channel topocentric spread to 0.798 channels, an rms of $0.351 \text{ channels} = 0.223 \text{ km s}^{-1}$ and a factor 24 improvement, across 4 epochs spanning 44 d in 488 kHz channels. It also places β Pictoris' CO(2–1) at $+3.52 \text{ km s}^{-1}$ in its 244 kHz windows against $+3.55 \text{ km s}^{-1}$ in its 15.3 kHz ones. The two agree to 0.087 km s^{-1} across a factor 16 in channel

TABLE 5

THE LEDGER IN COUNTS, EACH AGAINST WHAT CHANCE WOULD GIVE. THE LINE-MASK EXPECTATION IS THE MASK'S OCCUPANCY OF THE SEARCHED BAND TIMES THE 56 CROSSINGS. THE CHANCE EXPECTATION FOR AN UNATTRIBUTED CLASS A CROSSING IS SUMMED OVER EACH WINDOW'S OWN MEASURED FALSE-ALARM PROBABILITY (§5.3.1).

Quantity	Observed	Chance
Threshold crossings	56	—
Fitted in the visibilities	51	—
Untested (fit outstanding)	5	—
Line-attributed ($ \Delta v \leq 50 \text{ km s}^{-1}$)	16	2.3
Unattributed	40	53.7
Localised, committed epoch	11	0.35
Localised, adopted epoch	41	0.09
epoch residual verified	39	—
provisional ([†] , residual bounded)	2	—
of those, unattributed	31	—
Rank-screen flagged	12	—
Unattributed and rank-flagged	2	—
Unattributed, rank-flagged, recurring	0	—
Crossings with a recurrence test	10	—

width.

The gain is $\sqrt{N_{\text{eff}}}$ and not $\sqrt{N}$, where $N_{\text{eff}} = \sigma_{\text{min}}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs. The median N_{eff}/N is 0.65 and the tenth percentile 0.30. Measured against $\sqrt{N_{\text{eff}}}$ the realised improvement is 0.992, so the combination is ideal to about one per cent. Against $\sqrt{N}$ it is 0.78, and the shortfall is unequal epoch depth rather than registration error. An epoch count alone is therefore misleading, and N_{eff} is quoted with it throughout: HD161868 stacks 13 epochs for a gain of only 1.15, because its N_{eff} is 1.7. Carrier-equivalent, the median gain is 1.27 after the measured sub-channel registration loss of 1 – 0.875. Every stacked limit also carries a realisation term, because the noise scale is a median absolute deviation over a median of 127 common channels. That scatter is ± 7 per cent across the 368 groups, and it is irreducible at this epoch count.

The deepest stacked limit is *Proxima Centauri's*. *Proxima* holds blocks over h spanning yr, but its N_{eff} is only $\times$, and it is the second number that sets the realised gain of $\times$. Carriers injected into every epoch of that stack and recovered through the same path give a 90 per cent recovery point at $\times$ the stack's nominal threshold. The limit is therefore $\text{EIRP}_{90} = W$ for an emitter steady in the stellar frame to km s^{-1} across epochs spanning yr, with no inter-epoch drift. That is about times below this survey's per-system median, and as far as we are aware it is the deepest millimetre technosignature limit measured on any star through a recovery campaign. The calibration factor is a property of the stacking chain rather than of this star: a second group returns $\times$, and the factor is the ratio of the two thresholds, times the channel response of a carrier at a random sub-channel position, times the margin a ninety-per-cent point needs over a fifty-per-cent one. *Proxima's* stack is built from coarse 15.625 MHz windows, which is why its registration tolerance is so wide: it is robust to planetary reflex motion and dilute in bandwidth, the opposite trade from the fine-channel stacks.

groups pass the full criterion and both are known lines. They are β Pictoris' CO $J = 1-0$ at $+3.09 \text{ km s}^{-1}$ over 8 epochs ($Z = 45.3$) and its $J = 2-1$ at $+3.55 \text{ km s}^{-1}$ ($Z = 12.1$). The two bands agree to 0.46 km s^{-1} , or 0.73 of a CO(1–0) channel. That is the astrophysical posi-

tive control passing in the stack. Attributing every survivor against the survey's own $\pm 50 \text{ km s}^{-1}$ mask leaves 0 unattributed. The stack is also an independent test of the threshold crossings of §5.6: the 9 stars carrying them stack to $Z = 2.82-4.43$ in 8 cases, 0 exceed their own control ensembles and 0 passes the criterion. The exception is HD 285968, which reaches $Z = 12.89$ on the same frame-locked foreground line used above as the registration standard. It is rejected because its own controls reach 14.19: the emission is extended, which is what a foreground cloud does and a transmitter does not.

The restriction travels with every number above, and its width is set by the channel. The stack assumes an emitter whose velocity is steady in the stellar frame, and a channel is finite. At CO(2–1) a 488 kHz channel is 0.635 km s^{-1} , so a reflex motion of $\pm 30 \text{ km s}^{-1}$ — a transmitter on a short-period giant planet — is 23.1 MHz, about 47 channels. Such an emitter de-registers itself completely between epochs. The same applies to any intrinsic drift above roughly $1.3 \times 10^{-1} \text{ Hz s}^{-1}$ over the median inter-epoch span of 44 d, or $2.0 \times 10^{-3} \text{ Hz s}^{-1}$ over the longest (2868 d). Each stacked limit therefore constrains persistent emitters that are stable in the stellar frame to within its own channel, and nothing else. The single-epoch limits of §6.2 are the ones that apply generally.

5.8. What this experiment excludes, and what it does not

Excluded. A continuously transmitting, unresolved carrier lying outside the molecular-line mask, inside the searched frequency range and inside the linear-drift grid, and present during one of the archival epochs, would have been found nine times in ten above the EIRP_{90} of §5.1 — a median of $2.9 \times 10^{15} \text{ W}$ per system on its best window. No such carrier is present toward any of the 82 systems searched.

Not excluded. A transmitter absent during the archival observation, or on for less than about a tenth of the time: at a duty cycle of one tenth, recovery falls to 71 per cent for the fine class and 83 per cent for the coarse (Appendix ??), and those trials are undrifted, so the joint drifting-and-intermittent case is not measured at all. Emission inside the $\pm 50 \text{ km s}^{-1}$ molecular-line mask, at any strength. Drift outside the searched grid, or drift curved enough to leave a channel within the track. Signals broad enough to be removed with the continuum baseline. Anything below a window's own completeness. And any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

5.8.1. The survey's on-source time, in both directions

Two bookkeeping biases of opposite sign affect the exposure, and neither may be quoted without the other. The extractor retained a single FIELD row per execution block, so in 42 of the 404 released blocks — 171 windows toward 18 stars — further in-beam pointings on the same star were discarded, 19.1 h of real on-source time, 6.8 per cent of the survey. Nothing published is thereby wrong, because every noise level was measured from the data actually used; but the achievable sensitivity in those windows is understated by a median factor 1.73 in flux density, and those blocks were re-extracted

and re-searched. In the opposite direction, the released on-source time is the number of integrations times the median dump length rather than a sum of dump lengths, so short sub-scan boundary dumps are over-counted; the exact computation moves the survey total from 279.0 to 260.5 h. *The first bias costs the survey time it had and the second credits it with time it never had*, and they offset to within +0.6–+6.2 h.

5.8.2. The omitted annual parallax

The extraction phase-rotates the visibilities to each star’s *barycentric* direction while the visibilities see the apparent one, so annual parallax is omitted from the assumed phase centre. The resulting amplitude loss is the normalised dirty beam evaluated at the parallax displacement in units of the synthesised beam: parameter-free, multiplicative and shape-preserving. Over the 1654 windows carrying the astrometric keys it is 0.13 per cent in the median window and 51.5 per cent at worst, with 76 windows losing more than ten per cent. It is negligible throughout the ACA data that dominate this survey and matters only for nearby stars observed in long-baseline 12 m configurations. *We state it and do not apply it*. The sign is optimistic — every limit quoted here is too deep by $1/(1 - \text{loss})$ in its own window — and the completeness campaign cannot see the term, because its carriers are deposited at the same assumed position the estimator evaluates. Correcting it would leave the number of systems reaching 10^{15} W unchanged at 14, because a long-baseline block is never a system’s most sensitive window.

5.8.3. Two blind spots

The search is Stokes *I* only. A fully polarised carrier is not missed, but the discrimination that polarisation would offer against unpolarised astrophysical emission is unavailable. ALMA’s feeds are linear, so a per-hand search would help only a carrier aligned with one feed; a circularly polarised signal divides equally between XX and YY, and its circularity is in the cross-hands, which this analysis does not extract. The frequency coverage, finally, is the archive’s and not a choice. None of the “magic frequency” arguments that have guided centimetre-band searches — the 1420 MHz line, its multiples, the water hole, or π and e times them — falls in an ALMA band, so a transmitter deliberately placed on one lies outside this experiment by construction and its absence here is not evidence against it.

6. DISCUSSION

6.1. The search space this archive opens

The frequencies are what is new here. The windows searched span 118.1 GHz of unique sky frequency, of which 47.7 GHz carries channels fine enough to resolve a drifting carrier. Only 20.6 GHz of that lies inside the 10 MHz–115 GHz interval Wright et al. (2018) take as the frequency axis of the search haystack, the ionospheric cutoff to the top of the Green Bank Telescope’s instrumentation. The remaining 97.5 GHz in Bands 4–10 has not been searched toward nearby stars before. The targets of the one prior search of ALMA archival data (Mason et al. 2025) lie four orders of magnitude further away, so the comparison is best made in minimum detectable *received* flux at a matched channel width. On that footing the windows here reach 5.2×10^{-24} W m^{−2} at best,

a factor 9 deeper, and 7.1×10^{-23} W m^{−2} at the median, a factor 1.5 shallower.

There are physical reasons to look at millimetre wavelengths rather than merely to accept them as an archival accident. A transmitter’s directive gain rises as the square of frequency at fixed aperture, $G = \eta(\pi D\nu/c)^2$. A dish of modest size is therefore an extremely narrow, high-gain link in this regime: the 12 m aperture used as this paper’s benchmark has $G = 5.9 \times 10^8$ at 230 GHz. The absolute bandwidth available also scales with carrier frequency, which is what a high-rate data link or a wideband radar needs. Hardware that beams energy, for power transmission or propulsion or ranging, radiates in the same regime. Propagation is favourable as well. Interstellar dispersion and scattering both fall steeply with frequency, so a millimetre carrier arrives with far less temporal smearing and frequency-dependent delay than a centimetre one. The narrow beams that come free at high frequency also waste little power on empty sky. None of this makes millimetre wavelengths the likeliest place to transmit. It makes them a plausible one that had not been examined.

6.2. What this experiment excludes, and what it does not

Excluded. Consider a continuously transmitting, unresolved carrier that lies outside the molecular-line mask, inside the searched frequency range and inside the linear-drift grid, and that is present during one of the archival epochs. Such a carrier would have been found nine times in ten above the EIRP₉₀ of \$5.1, a median of 2.9×10^{15} W per system on its best window. None is present toward any of the 82 systems searched.

One benchmark carries the physical reading of that figure. A 12 m dish radiating 1 MW at 230 GHz, at an aperture efficiency of 0.7, reaches an EIRP of 5.9×10^{14} W, and the survey is sensitive to that depth for 11 of the 60 Class A systems. *EIRP is not transmitter power*. The same megawatt radiated isotropically would be 10^6 W of EIRP, and it is the directivity, a factor 5.9×10^8 here, that lifts a modest transmitter into the range this survey probes. The exclusion therefore applies to beamed transmitters whose beam contains the Solar System, and says almost nothing about omnidirectional beacons.

Four further axes bound the excluded population as sharply as power does. *Frequency*: the coverage is sparse and inherited, 47.7 GHz of unique Class A frequency in a window function set by other astronomers’ science, and the median system is covered over only 1.84 GHz. *Drift and acceleration*: the searched grid spans 12.0–13.3 Hz s^{−1} GHz^{−1}, a line-of-sight acceleration of 3.60–4.00 ms^{−2}. That contains the rotational and orbital accelerations of the 50 known planets of the 23 hosts in the sample, only TRAPPIST-1 b reaching it in its edge-on worst case. A transmitter whose own oscillator drifts faster than that kinematic budget lies outside the grid. *Persistence*: the carrier must be radiating during one of the archival observations, so these are limits on persistent emission. *Spectral width*: the statistic is built for emission confined to one 15.3–1953 kHz channel. A carrier broadened beyond a channel loses signal to its own width, and a signal broad enough to resemble continuum is removed with the spectral baseline.

Not excluded. A transmitter absent during the archival

observation, or on for less than about a tenth of the time. At a duty cycle of one tenth, recovery falls to 71 per cent for the fine class and 83 per cent for the coarse (Appendix ??); those trials are undrifted, so the joint drifting-and-intermittent case is not measured at all. Emission inside the $\pm 50 \text{ km s}^{-1}$ molecular-line mask, at any strength, which costs 4.81 GHz of coverage. Drift outside the searched grid, or drift curved enough to leave a channel within the track. Signals broad enough to be removed with the continuum baseline. Anything below a window's own completeness. And any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

6.2.1. The survey's on-source time, in both directions

Two bookkeeping biases of opposite sign affect the exposure, and neither may be quoted without the other. The extractor retained a single FIELD row per execution block. In 42 of the 404 released blocks — 171 windows toward 18 stars — further in-beam pointings on the same star were discarded, 19.1 h of real on-source time and 6.8 per cent of the survey. Nothing published is thereby wrong, because every noise level was measured from the data actually used. But the *achievable* sensitivity in those windows is understated by a median factor 1.73 in flux density, and those blocks were re-extracted and re-searched. In the opposite direction, the released on-source time is the number of integrations times the median dump length rather than a sum of dump lengths, so short sub-scan boundary dumps are over-counted. The exact computation moves the survey total from 279.0 to 260.5 h. *The first bias costs the survey time it had and the second credits it with time it never had*, and they offset to within $+0.6$ – $+6.2$ h.

6.2.2. The omitted annual parallax

The extraction phase-rotates the visibilities to each star's *barycentric* direction while the visibilities see the apparent one, so annual parallax is omitted from the assumed phase centre. The resulting amplitude loss is the normalised dirty beam evaluated at the parallax displacement in units of the synthesised beam: parameter-free, multiplicative and shape-preserving. Over the 1654 windows carrying the astrometric keys it is 0.13 per cent in the median window and 51.5 per cent at worst, with 76 windows losing more than ten per cent. It is negligible throughout the ACA data that dominate this survey and matters only for nearby stars observed in long-baseline 12 m configurations. *We state it and do not apply it*. The sign is optimistic: every limit quoted here is too deep by $1/(1 - \text{loss})$ in its own window. The completeness campaign cannot see the term either, because its carriers are deposited at the same assumed position the estimator evaluates. Correcting it would leave the number of systems reaching 10^{15} W unchanged at 14, because a long-baseline block is never a system's most sensitive window.

6.2.3. Two blind spots

The search is Stokes *I* only. A fully polarised carrier is not missed, but the discrimination that polarisation would offer against unpolarised astrophysical emission is unavailable. ALMA's feeds are linear, so a per-hand

search would help only a carrier aligned with one feed. A circularly polarised signal divides equally between XX and YY, and its circularity is in the cross-hands, which this analysis does not extract. The frequency coverage, finally, is the archive's and not a choice. None of the “magic frequency” arguments that have guided centimetre-band searches — the 1420 MHz line, its multiples, the water hole, or π and e times them — falls in an ALMA band. A transmitter deliberately placed on one lies outside this experiment by construction, and its absence here is not evidence against it.

6.3. What the archival experiment taught us

Molecular-line confusion, and not interference or instrumental artefacts, is the dominant obstacle to a millimetre technosignature search. Catalogued molecular transitions, almost all of them carbon monoxide, account for 14 of the 56 threshold crossings, and 10 of the 12 windows that reached the spatial stage are disc or foreground CO, arising in 2 of the 69 debris-disc systems searched — one flagged system in 34. The remedy that works is a velocity-space mask evaluated in the stellar frame, and it is not free: it removes 4.1 per cent of otherwise usable coverage. The spatial screen is weakest in the same place. A line that fills the primary beam raises the star and the control annulus together instead of cancelling, so the screen loses power exactly where circumstellar gas is brightest. A future millimetre search inherits that confusion rate. It should measure the continuum at each target before promoting a window, rather than lowering a threshold toward astrophysically interesting targets.

Repeat epochs are what turn a crossing into a candidate, and an archive supplies them only incidentally. Temporal sampling is the weakest axis of this experiment, and for close to half the sample it constrains an intermittent emitter hardly at all, whatever its power. Archival coverage is also incomplete at the level of whole observations. 144 of the 177 execution blocks in scope were searched. 18 carry no ALMA pipeline calibration at all, 11 failed calibration, 1 is infeasible at this data volume and 3 were not attempted. For one of them the observatory delivered raw data it had never calibrated, so part of that gap lies in the delivery rather than in the processing.

Completeness has to be measured rather than inferred from a threshold, and measuring it is expensive. A nominal 5σ level is a property of the noise. It is not a statement about what a transmitter of a given power would do to a pipeline, and the two differ here by the factor 5.70 in Class A and 4.55 in Class B. The cost is that each injected window must be re-extracted and re-searched over its full drift grid once per amplitude. That is why the campaign covers a sample of windows and transfers its ratio to the others. Two limitations of a different kind should be read with every limit. The null result rests on internal validation alone: every check reported here tests this pipeline against itself. Because the only prior ALMA-archival search targets a disjoint sample, there is no external pipeline to compare against. And the threshold crossings are not concentrated toward window edges. An apparent pooled excess there comes almost entirely from one window carrying resolved CO close to the band edge, and both removing that window and masking the line leave the ratio consistent with unity.

Archival SETI buys sky and frequency cheaply and buys confirmation not at all. This experiment reached 82 systems and 118.1 GHz without an allocation, at a sensitivity no targeted programme at these frequencies has matched. It did so on data selected by others. 90 per cent of the searched stars were observed for disc or planet-formation science, which puts circumstellar CO in a large fraction of the fields, and only 21 of the 5 908 catalogued M dwarfs within 40 pc have any coverage at all. The quantities above therefore describe *these* stars, frequencies and epochs and nothing wider. The non-detection cannot be converted into an occurrence rate for transmitters within 40 pc, which would need a denominator an archive-selected sample does not supply. On the conventional Continuous Waveform Transmitter Figure of Merit this survey is some four orders of magnitude less competitive than major centimetre-wave programmes. That figure measures the frequency regime and the archival temporal sampling rather than the design. Comparison is in general better made axis by axis, because this survey differs from centimetre-wave searches in frequency, spectral resolution, number of systems, completeness, drift range and temporal coverage at once, and those axes are coupled.

6.4. *What a dedicated ALMA technosignature experiment should do differently*

A purpose-built experiment on the same telescope would be a far stronger one, and the changes it needs follow from the limitations above. *Channelise finely everywhere:* 1249 of the 1651 windows searched here have channels so coarse that every in-grid drift is sub-channel over a track. They measure amplitude but cannot distinguish a drifting carrier from a stationary one. *Choose the targets:* an archive weighted toward dusty discs is the worst case for molecular-line confusion, and the M dwarfs that dominate the nearby habitable-zone planet census are the least covered. Both are decisions a proposal can simply reverse. *Choose the frequencies:* windows placed deliberately away from the bright molecular transitions would recover most of the 4.81 GHz the mask costs. *Repeat the epochs on purpose:* two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity. *Keep contemporaneous off-source controls:* full dynamic spectra survive for only 8 of the 512 control positions here, which is what limits the second-stage null in a structured field. Retaining them all costs storage and nothing else. *Record polarisation, including the cross-hands:* a per-hand search costs no sensitivity and buys discrimination against weakly polarised astrophysical emission, and only a cross-hand extraction can see a circularly polarised carrier. *Retain the visibilities:* rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility, with an imaginary part consistent with zero on every baseline. That asks whether the data contain a source *where the star is*. It is a physically specific question, and a stronger one than any rank against neighbouring positions in an image.

7. CONCLUSIONS

1. ALMA's public archive opens a region of the technosignature search space that had not been examined

toward nearby stars. Of the 118.1 GHz of unique sky frequency searched here, only 20.6 GHz lies inside the frequency interval covered by centimetre-wave programmes.

2. The primary experiment is 402 fine-channel spectral windows toward 60 stellar systems, covering 47.7 GHz of unique frequency, searched for an unresolved carrier whose frequency drifts. A secondary coarse-channel spectral-excess search covers a further 70.4 GHz in 1249 windows but cannot discriminate drift, and both come from one uniform reduction of 484 public execution blocks, 4.1 TB of raw data.
3. The sensitivity is measured by injection and recovery through the complete automated selection rather than inferred from a threshold: $\text{EIRP}_{90} = 5.70 P_{\text{trig}}$, a median of 2.9×10^{15} W per system. That is deep enough for 11 of the 60 Class A systems to exclude a 12 m dish radiating 1 MW at 230 GHz, whose EIRP is 5.9×10^{14} W.
4. 56 threshold crossings are reported with their dispositions, and 14 are attributable to catalogued molecular transitions, almost all of them circumstellar or foreground carbon monoxide, among them the β Pictoris debris disc recovered blind in two transitions. None of the remaining 40 is independently confirmed: none recurs in the repeat observations available, and 2 also outrank all 512 spatial controls.
5. Combining the epochs of each star in that star's rest frame deepens the flux limit by a median factor 1.46 over the best single epoch. For Proxima Centauri the stacked limit, measured by injection into the stack, is W for an emitter steady in the stellar frame to km s^{-1} across epochs spanning yr.
6. These results constrain persistent, unresolved carriers within the frequency, drift, duty-cycle and sensitivity domain actually searched, toward the stars ALMA happened to observe. They cannot be converted into an occurrence rate for transmitters in the 40 pc stellar population, which would require a denominator an archive-selected sample does not supply.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection-recovery campaigns, and produced the released catalogue and its regeneration scripts. Both authors designed the candidate vetting procedure, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of Table 4.

FUNDING

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COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the

selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 404 execution blocks searched span 2013 October 6 to 2025 June 11 across 63 proposal codes, which in the standard ADS/JAO.ALMA#*code* form are: 2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S. Per-block epochs are in the machine-readable release. The archive metadata exposes no proposal code for 2 of the 404 blocks (A002_X11d9ce7_X5257, A002_Xbf792a_X26ec), which are therefore not represented in that list. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France).

DATA AVAILABILITY

The release carries every reduced data product, the full target-selection and analysis pipeline, the per-window catalogue, the injection records and the control statistics. *The catalogue is the 1651-row release of the previous version*; the repaired search statistic adopted in §?? is deposited beside it, one row per re-extracted window, together with the frozen records behind Appendix ?? — the harvest, the matched recurrence fits, the clustering recomputation, the archival tail’s own verdicts and the ALMA delivery record for the block the ledger had left unnamed. What has been demonstrated, rather than claimed, is narrower and is stated in Appendix ??: 125 generated products regenerate byte-identically from the frozen inputs, and 41 headline quantities are re-derived from the released catalogue alone, without the pipeline. *The reference of record is the Zenodo deposit*, which is frozen, carries a DOI minted on acceptance, and is the

version any reproduction should cite; the repository at <https://github.com/Tilanthi/SETI> is a working mirror and may move on.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

A reader checking a threshold in this paper needs three things: the frame every frequency is quoted in, the instrumental response that separates a channel threshold from a transmitter power, and the primary-beam factor that converts a measured noise into a flux at the star. They are collected here. The per-window catalogue, `per_target_results_v3.99.csv`, carries one row per searched window and the columns each of these corrections uses.

Frequency frames, linewidth and smearing

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, with no velocity-frame shift applied at any stage. Velocities enter only in the vetting stage’s line mask, which is evaluated in the same observed sky frame (Appendix C): all 1396 released `line_offset_MHz` values reproduce topocentrically against the frozen rest-frequency list. The tolerance is wide enough to absorb the barycentric and systemic terms it does not apply, so the residual error is the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. Channel dilution spreads a sub-channel carrier’s total flux across the channel, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, independent of the transmitter’s own width below the channel, and never a spectral power density. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1 commensurable. One number fixes the scale, illustratively and not achievably: the median Class A channel is 488 kHz, so rechannelising the same data to the 2.79 Hz of a dedicated narrowband search would, on $P_{\text{min}} \propto \sqrt{\Delta\nu}$, divide the threshold by 418 and put the median Class A selection power at $\sim 7.4 \times 10^{12} \text{ W}$. ALMA cannot do this: the correlator delivered the channelisation the archive holds, and no reprocessing recovers resolution that was never recorded.

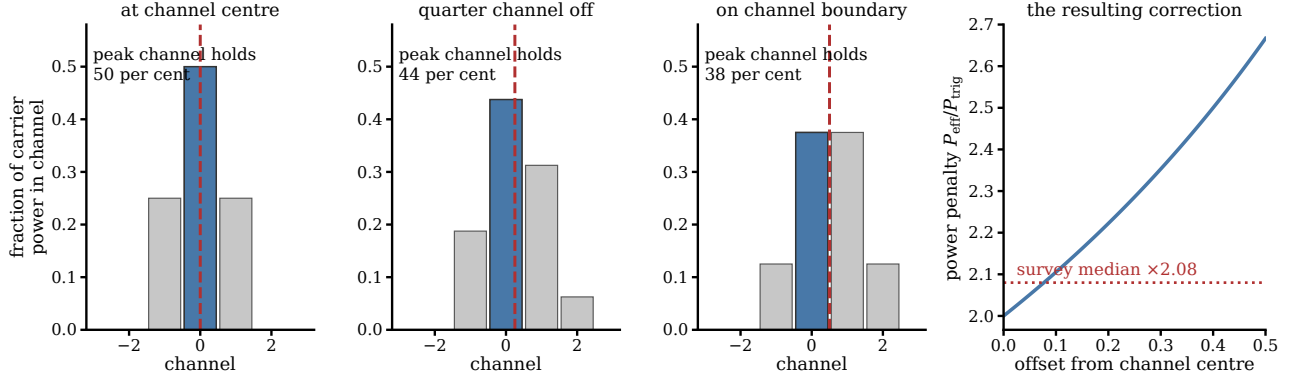


FIG. 6.— Why the nominal channel threshold is not the transmitter threshold. ALMA applies online Hanning smoothing by default, so an intrinsically unresolved carrier is spread over neighbouring channels and the peak channel holds only part of its power: half at a channel centre, 0.42 on a boundary (first three panels; blue is the peak channel, the dashed line the carrier). The search threshold is set in units of the channel noise, so the power a transmitter must radiate to reach it is larger than the nominal figure by the reciprocal of that fraction (right-hand panel). This is the whole of the difference between P_{trig} and P_{eff} .

Channel widths come from the measurement sets’ own spectral window tables (CHAN_WIDTH). A missing “EFFECTIVE_BANDWIDTH” column is not evidence of an absence of online spectral averaging, MS v2 defining no such column, so the quoted channelisations are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default, so the spectral response is ~ 2 channels wide and neighbouring channels are correlated (Fig. 6). For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak-channel fractions 0.42–0.50 that follow, median 0.48, and the resulting $\times 2.00$ – $\times 2.36$ optimism of the nominal thresholds are given in §4; the released catalogue carries the nominal threshold and both corrected values per window, and the sensitivity summary quotes the worst case. The injections deposit an unsmoothed delta-in-channel tone, so the recovery curves carry the same bias and the end-to-end check cannot see it.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null have a measured lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so that correlation is instrumental and the residual spectra are smoothed and un-averaged. And the archive’s reported effective spectral resolution per window, against each measurement set’s own channel separation, is exactly 2.00 in 1505 of the 1651 windows, in both correlator modes and including 1217 of the 1249 coarse time-division-mode ones; the other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller, and the 0 Band 9 and 10 windows, absent from the metadata snapshot, take ALMA’s default. The blanket factor is right for the great majority of both classes. Thresholds are not rescaled: each is nominal and up to $\times 2.36$ optimistic. *The correlation model is truncated at lag two*, and the baseline step — a per-integration median in blocks of up to 65 channels — could imprint correlation on longer scales, which the released products retain no residual spectra to measure. Such a term would make the independent-trials count of Appendix D an over-estimate and the achievable per-window resolution coarser still.

Linewidth convention. The statistic is the single-channel flux density after drift correction, so the quoted $\text{EIRP}_{\text{trig}}$ applies to any transmitter narrower than the native channel and is flat in the assumed linewidth. Two sub-channel qualifications attach: flatness is exact only for a centred tone, a boundary-straddling one splitting its power between two channels at worst by a factor of two; and the channel-to-channel transition follows the correlator’s passband shape, which the frozen products do not retain. Thresholds therefore carry a sub-channel-position systematic of tens of per cent, and EIRP values are quoted to two significant figures.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is ≈ 32 MHz at 488 kHz and ≈ 0.9 GHz at 15.62 MHz. Within that ceiling, a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so to first order

$$\text{EIRP}_{\text{trig}}(N_{\text{ch}}) \simeq N_{\text{ch}} \times \text{EIRP}_{\text{trig}}(1), \quad N_{\text{ch}} \lesssim W/2, \quad (\text{A1})$$

the rescaling the main text applies to signals broader than one channel.

Intra-integration smearing. The frequency excursion at the maximum searched drift rate across one integration, $\dot{\nu}_{\text{max}} t_{\text{int}}$, is at most 1.10 channels across the 1651 windows (median 0.001, 90th percentile 0.06). A linear drift smeared by Δ channels within an integration keeps a fraction $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak-channel amplitude, so the effective threshold is $\text{EIRP}_{\text{trig}}/\eta_{\text{smear}}$; a pure occupancy argument would retain $\min(1, 1/\Delta)$ instead, giving 1.0 rather than 0.64 at $\Delta = 1$, so the sinc form is the conservative reading. The 9 windows with $\eta_{\text{smear}} < 0.99$, as star, band, (η_{smear} , threshold multiplier), are HIP 10679 Band 6 (0.574, $\times 1.74$), β Pic Band 6 (0.574, $\times 1.74$), HD 172555 Band 6 (0.574, $\times 1.74$), HIP 10679 Band 6 (0.606, $\times 1.65$), HD 172555 Band 6 (0.606, $\times 1.65$), η Crv Band 8 (0.865, $\times 1.16$), η Crv Band 7 (0.932, $\times 1.07$), HD 48370 Band 6 (0.979, $\times 1.02$) and CP–72 2713 Band 6 (0.979, $\times 1.02$); the remaining 442 agree with their nominal thresholds to bet-

ter than 1 per cent. Compounding the worst smearing penalty with the worst Hanning factor gives $\times 4.6$ on the nominal threshold for those five windows, and no released column carries that product.

The primary-beam response, and the two columns that let a reader recompute it

The sensitivity quoted for a window is $S_{\min} = 5\sigma/A$, where A is the primary-beam response at the star's offset from the phase centre: the noise σ is measured in apparent flux, the limit is quoted in true flux, and the two differ by A . That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis: over the 1651 released windows the offset has a median of $0.089''$, a 99th percentile of $7.43''$ and a maximum of $8.13''$, and A runs from 0.999984 in the median down to 0.5522, below 0.99 on 91 windows and below 0.9 on 34. *None* falls below one half, and every one of the low-response windows is a physically explicable J2000 pointing on a fast-moving star rather than an error. A window is kept only where the response is at least 0.5, a correction of at most $\times 2$, since at low response a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap: the last retained window has response 0.55 at $0.46\theta_{\text{PB}}$, the first window beyond it 0.35 at $0.62\theta_{\text{PB}}$, and nothing lies between, so the rule binds prospectively and excludes none of the 1651 released windows. Beyond it are the 12 ϵ Eri Band 6 windows at $1.0\text{--}3.5\times$, also past the first sidelobe transition where the Gaussian form fails; the two criteria are independent and select the same windows. They are *withheld*, enter no number here and are absent from the released catalogue, although under the nominal rule they would have covered $3.7 \times 10^{13}\text{--}1.1 \times 10^{15}$ W at 3.22 pc.

The correction is applied exactly once, and that is the complete correction. The offset is computed for every measurement set from its own FIELD:PHASE_DIR and the propagated stellar position, so no window carries a zero-by-assumption offset. It enters the published numbers in one place only: the identity $S_{\min} = 5\sigma/A$ closes on all 1651 released rows to $8.7\text{e-}05$ in relative terms. *The count is given in two parts, because on part of it the identity cannot fail:* on 10 rows whose products are not retained A is back-solved from $5\sigma/S_{\min}$ and is flagged as such in the catalogue, so there the closure is arithmetic and shows it, at $4.0\text{e-}07$ against $8.7\text{e-}05$ on the 1641 rows where A comes from the search's own stored geometry. 1641 is therefore the number of rows on which the audit is a test. Nothing else needs correcting, because the detection statistic of Eq. 1, the control ranks and the attribution are all *ratios* formed inside a single window, in which A divides out identically; no crossing, no rank and no disposition depends on it.

Two catalogue columns, `pb_offset_arcsec` and `pb_atten`, carry the offset and the response per window so that a reader can repeat the calculation. They are needed, because the only other geometry column, `theta_pb_arcsec`, is the pipeline's $1.22\lambda/D$ evaluated at $D = 12$ m for *every* window — including the 1054 rows taken on the $1.7\times$ smaller ACA dishes, where it is correspondingly too narrow — so recomputing A from it alone would give the wrong answer on the majority of the cat-

alogue. The annulus geometry keeps the pipeline's own $\theta_{\text{PB}} = 1.22\lambda/D$, an Airy first-null coefficient; recomputing it on ALMA's FWHM $1.13\lambda/D$ at each window's own dish diameter moves 75 of 1651 windows by more than 1 per cent, extremes -2.6 to $+10.3$ per cent, and no headline threshold moves.

The beam model itself is a bounded systematic, and it is bounded by measurement. `pb_atten` reproduces the Gaussian at $1.22\lambda/D$ that the pipeline applied; CASA's blocked-Airy response is the better physics. Evaluated window by window, the largest ratio between the two over the whole release is 1.0745, and the counts of systems reaching 10^{15} W are identical under either model.

The injection campaign is blind to A by construction, and the resulting bias is exactly zero. The injector adds an unattenuated amplitude at the star's own direction cosines, so no primary-beam factor enters the signal it deposits. That would bias the measured P_{90}^{sel} if the ladder were calibrated in true flux — and it is not. On all 40 scored units the trigger flux the ladder is referred to is the *apparent* 5σ , so an injected apparent amplitude a recovered nine times in ten implies a true flux a/A , while P_{trig} is the power corresponding to a true $5\sigma/A$, and A cancels from the ratio identically. 4 of the 40 units have apparent and corrected $S_{\min}$ differing by more than a part in a thousand, so the test distinguishes the two conventions and is not vacuous. Two limitations remain, and they are limitations rather than corrections. First, *this campaign can never falsify the beam model*: being self-consistent in apparent flux by construction, no injection outcome can disagree with A . Second, injected configurations span $0.011''$ to $4.98''$ (median $0.033''$, $A \geq 0.9102$), whereas 75 released windows on 5 stars lie at larger offsets, reaching $A = 0.552$; for those the completeness curve is *interpolated rather than measured*. Their 6 crossings are unaffected, being ratios, but a limit quoted for one of those stars rests on an extrapolated P_{90} .

Four rows carry a published noise that no retained product reproduces, and they are declared rather than reconciled: 4 windows of HD 285968 in execution block A002_Xd98580_X3bf6, whose published σ is $0.88\text{--}1.32$ per cent *higher* than the surviving product's, so the published limits there are weaker and not stronger. With the 10 windows whose products are not retained at all, the declared provenance gap is 14 windows. The two attenuation columns are unaffected: for these rows A comes from a retained product at the same position, where $A \geq 0.999960$, and $S_{\min} = 5\sigma/A$ still closes to $2.6\text{e-}05$.

THE SPATIAL SCREEN: CALIBRATION AND VALIDATION

The spatial screen asks whether an excess sits at the star or anywhere in the field, and it removes most of the unattributed crossings. Everything the dispositions rest on therefore depends on how well that screen is calibrated, which is what this appendix measures. The answer is adverse to the screen and is stated as such: The star-versus-control rank is an empirically useful candidate-prioritisation statistic, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and that asymmetry has a known instrumental mechanism. Images are corrected for the primary-beam response, so the true noise rises with distance from the pointing centre while the search divides every position by one scale estimated over the whole window; the control probes therefore sit at systematically different effective noise from the star. A radius-corrected normalisation removes that term: each control probe is standardised within its own radius bin, the probe radii being known from the fixed draw, with no free parameter.

Applied out of sample, the correction does not repair the statistic. It was frozen — probe radii, bin edges and standardisation alike — and applied unmodified to 778 windows that played no part in its construction. The uncorrected stellar rank has median 0.409 against the exchangeable value of 0.5; after correction the median is 0.407, moving further from it, and the result holds separately in both out-of-sample partitions (hold-out 0.405→0.392, extension 0.409→0.417). Rescoring the whole survey with the gradient removed takes the rank-passing list from 12 windows to 11, losing 2 and gaining 1, all marginal and all dispositioned on other grounds, and the median rank over the 1614 windows that retain a control vector moves from 0.485 to 0.493, slightly further from 0.5.

The interpretation is narrow and, we think, unavoidable. The radial gradient is real and is one mechanism behind the displacement, but removing it does not leave an exchangeable statistic, so the residual is not radial. Neither normalisation yields a calibrated tail probability. **We therefore retain the frozen screen as primary — it has at least the virtue of having been fixed in advance — use it only to decide which windows are examined, and rest every disposition on physical evidence or on recurrence rather than on either rank.** The radial correction, the region-maximum audit and the pseudo-star tests each leave every disposition unchanged.

The reserved hold-out, and what it calibrates

The statistic and the mask were settled on the searched sample and cannot also calibrate themselves, so a hold-out was reserved by a rule fixed and recorded before the archival search finished and before any reserved block was searched: a block is held out if and only if $\text{sha256}(\text{UID})_{[1:8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — a block already in a previous release stays in the survey, and no system loses every block — so that the assignment can be recomputed from the UIDs alone. The three choices that can manufacture a candidate were fixed before the hold-out rule itself: the detection statistic of Eq. 1, the promote and retire criteria, and then the reservation, with the first reserved block searched 3 days later. The diagnostics built afterwards — the radius-corrected normalisation, the visibility-domain test, the drift-resolved injection analysis and the P_{90} budget — cannot promote a window.

Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. Three quantities are calibrated on those blocks and on nothing else.

First, the rank displacement. The stellar add-one rank

out of sample has median 0.405 where 0.5 is expected, with a block-clustered 95 per cent interval of 0.341–0.458 over 315 windows, and 14 per cent of ranks below 0.1 against 6 per cent above 0.9, so the distribution is displaced rather than growing a tail. Clustering removes neither the shift nor its size: 99 of 149 blocks have a mean rank below 0.5, and a star-clustered bootstrap places the median at 0.440 (95 per cent 0.42–0.46). The searched sample leans the same way without resolving it (median 0.49, star-clustered 95 per cent 0.47–0.51), and the two distributions are indistinguishable, so the in-sample checks were underpowered rather than contradicted. A displacement that survives a hold-out chosen before the data were seen is a property of the statistic and not of this sample, which is the quantitative reason the rank is a screen and never a significance.

Second, the false-alarm tail factor — the measured rate at which a no-signal position ranks first, divided by the rate exchangeability would give. It is 1.2 on 5 040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9. That factor and its width set the accuracy of every chance expectation in this paper. The reserved blocks contain 0 rank-passing crossings against 0.61 expected.

Third, the radial behaviour of the control ensemble. The standardised control level runs +0.12, +0.10, +0.12, +0.16 from the inner to the outer edge of the annulus, a mean per-window slope of $+0.067 \pm 0.033$: the displacement reproduces out of sample, but the steep inner-edge excess seen in sample does not — which is why the radial term is one demonstrated mechanism for the displacement rather than the whole of it.

A second out-of-sample set is used in this paper and must not be conflated with the hold-out. The *external calibration sample* is 326 execution blocks toward 56 stars, reserved by a rule fixed before the archival search finished, and it is not a subset of the in-scope blocks of §3: it includes blocks toward stars outside the 40 pc work list, so its block count cannot be placed inside that ledger and enters no count of the science sample. It is used only to calibrate the rank statistic, and it carries the repeat observations the hold-out does not. Over its 1322 windows the frozen pipeline returns 7 rank-passing crossings, of which 4 are β Pictoris CO — the positive control recovering on data it had never seen — and 3 are unattributed, against 3.7 expected at the measured tail rate.

One crossing through the whole chain

CP–72 2713 is a worked example of the vetting chain rather than a result, and it is the clearest one the survey contains because every stage returns a verdict. At the chance expectation of Appendix D, 0.89 such events are expected over the Class A windows and 2 are observed.

In Band 7 the crossing reaches $T_\star = 5.809$ and outranks every one of the 512 control positions, so it passes the spatial screen; it is channel 35 of its window; and no transition of the frozen mask lies within $\pm 50 \text{ km s}^{-1}$ of it, so it is unattributed. CO(3→2) falls in this same window and is absent, below a 5σ limit fainter than the crossing itself, and sulphur-bearing species are not in the mask and were not in it before the search. The drift-following visibility fit places the emission at the stellar position, $\text{Re}/\sigma = 5.1027$ with $\text{Im}/\sigma = 0.7558$ against a

TABLE 6

BAND-BY-BAND COST OF THE MOLECULAR-LINE EXCLUSION.
 UNION: UNIQUE SKY FREQUENCY SEARCHED IN THE BAND.
 MASKED: THE PART OF IT INSIDE A $\pm 50 \text{ km s}^{-1}$ STELLAR-FRAME
 TUBE AROUND A CATALOGUED TRANSITION OF ONE OF THE
 MASKED SPECIES. LEFT: WHAT A TECHNOSIGNATURE SEARCH CAN
 STILL OCCUPY. THE MASK IS EXCLUDED SEARCH SPACE AND NOT
 A VETO (§??); THE LOSS IS CONCENTRATED IN BAND 6, WHERE
 $\text{CO } 2 \rightarrow 1$ SITS.

Band	Union (GHz)	Masked (GHz)	Left (GHz)	Lost (%)
3	20.64	0.272	20.37	1.3
4	6.94	0.049	6.89	0.7
5	6.94	0.065	6.87	0.9
6	29.45	1.374	28.07	4.7
7	26.74	1.589	25.15	5.9
8	6.68	0.164	6.52	2.5
9	6.94	0.454	6.48	6.5
10	13.80	0.844	12.96	6.1
All	118.13	4.811	113.31	4.1

control-position maximum of 1.8222; by the argument of §5.5.1 that is a consistency check and not a promotion.

What disposes of it is the repeat block. That second block matches the first in tuning, channel width, channel count, integration count and on-source time, registers to 0.18 of a channel, and is the deeper of the two, so a persistent emitter at the first block’s flux would have returned $T_\star = 6.21$. Pinned to the first block’s own frequency and drift it returns 3.60, and read as two measurements of one constant amplitude the pair is inconsistent at 4.6σ . The blocks are 2.05 h apart, so this is a within-night test: it excludes a persistent emitter of that strength on that baseline and excludes nothing about an intermittent one. **The disposition rests on that recurrence test and not on the rank**, which is the chain this paper applies to every crossing it cannot attribute.

THE VELOCITY-SPACE LINE MASK, AND WHAT IT COSTS

Attribution decides the candidate list, and the mask decides attribution, so the mask’s definition and its price are both given here.

Definition. A one-channel coincidence test is too tight for circumstellar gas: a native channel spans a fraction of a km s^{-1} while real emission is shifted by many channels. The tolerance is set by the kinematic budget of the gas rather than by the instrument — Keplerian motion where these molecules survive ($\lesssim 13 \text{ km s}^{-1}$ beyond 10 au) and linewidths $\lesssim 1 \text{ km s}^{-1}$ — and is $\pm 50 \text{ km s}^{-1}$ around each catalogued transition. It is evaluated topocentrically, in the observed sky frame (Appendix A), which is wide enough to absorb the barycentric and systemic terms it does not apply. The result is empirically conservative: the largest attributed circumstellar component (β Pic, within 0.39 km s^{-1} of systemic) sits at 0.8 per cent of the half-width.

The mask holds 17 transitions of 11 species at full JPL/CDMS precision (Pickett et al. 1998; Müller et al. 2005). The species criterion is the one objective rule storable in advance: the transitions ALMA’s band definitions are built around — the CO isotopologues, HCN, HCO^+ , CS, SiO, CN, H_2CO and $[\text{C I}]$ — which dominate the emission of disc-selected fields.

The half-width cannot have been tuned to remove a candidate. Reclassifying all 56 crossings at ± 20 to $\pm 100 \text{ km s}^{-1}$ costs 2.3 to 8.7 per cent of gross bandwidth, and every crossing that narrowing the mask releases as

unattributed is an identified carbon monoxide line of this survey’s own: the two rank-passing unattributed crossings are the same pair at every half-width in that range. Widening the mask cannot suppress anything further. The species list is equally robust: the alternative selection of every Splatalogue transition over the 46 searched islands with $E_u \leq 150 \text{ K}$ and $\log A_{ij} \geq -5$ — 5,114 transitions of 235 species — masks 75.1 per cent of the union against 4.1 per cent here and changes no disposition, because no masked crossing exceeds its own control ensemble. “Coincident with a known molecular line” is therefore a statement about a catalogue rather than a binary property, and no plausible widening or narrowing of either the half-width or the species list moves a window.

The cost. Masking removes 4.1 per cent of the unique-frequency union and 4.7 per cent of window-summed gross bandwidth, CN and CO dominating. It is not uniform, so Table 6 gives it band by band: it reaches 6.5 per cent in Band 9 and stays below 1.5 per cent in Bands 3–5. Nor is it shared equally between the two experiments — 7.4 per cent of the Class A union against 3.1 per cent in Class B — so the fine-channel experiment pays 2.4 times the fractional price, its coverage being concentrated in the line-tuned windows a molecular mask is designed to remove. The two do not sum to the total because the classes overlap in frequency.

The masked crossings are not near misses, and the unmasked ones are not near hits. Of the 56 windows carrying an on-star crossing, 12 also beat their control ring, 10 of those lying inside the mask and 2 outside it; 6 more lie inside the mask without beating their ring, of which 3 are on HD 285968 alone at -46.7 to -34.4 km s^{-1} from $\text{CO } 2 \rightarrow 1$, each already rejected by its own ring. The remaining 38 are ordinary noise crossings whose own control ring reaches or exceeds the stellar value (ring-maximum median 5.82 against a crossing median 5.18). None of them sits close to the mask edge: the nearest lies 127 km s^{-1} from any catalogued transition of a masked species (LHS 1140, 2.5 times the adopted half-width), and 0 have no such transition in the window at all.

Masking is a design selection effect, and it has a price we cannot buy off. It separates discovery from classification: a carrier at a line frequency is as detectable as anywhere else, but it cannot be classified, because the astrophysical prior at a catalogued transition is overwhelming. Removing those frequencies from the searched domain is the conservative choice, at the cost that *we place no constraint at all on artificial emission lying inside the masked regions, at any signal strength.* A better design searches every frequency and attaches a *lower prior* to a line-coincident event instead, since these are exactly the frequencies a communicator wanting to be found near a natural reference might choose (Cocconi & Morrison 1959). The released channels support that second search, and the released products keep a known-line-coincident list: every on-source threshold crossing inside the masked intervals, with its offset from each nearby transition.

THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

The paper’s central claim is that the number of unattributed crossings is the number chance predicts. This appendix gives the expectation that claim rests on,

what it is conditioned on, and what the control ensemble that calibrates it can and cannot support.

What the expectation is conditioned on

The expectation is built from the control statistics themselves, which are the only empirical null the data provide, and it imposes the condition an event must actually satisfy. Under the disposition chain this paper applies, that condition is to reach the trigger and outrank every control. Drawing one control index per execution block — so that windows sharing weather, calibration and correlator setup are not treated as independent trials — and resampling blocks, the expectation over the 402 Class A windows is 0.89 unattributed rank-passing crossings against the 2 observed, and the probability of that count or more is $P(\geq 2) = 0.22$. Clustering changes the number little, because the *position* of a block's control maximum is not shared across its windows.

The conditioning is restrictive, and that is what makes it informative. Exactly one probe per window is the ensemble maximum, so the probability that the stellar position would have been that maximum, treated as exchangeable with the probes, is $1/(512 + 1)$ in windows whose control maximum reaches $T_{\text{trig}} = 5$ and zero in the rest. Only 447 of the 1614 windows that retain a control vector have a control maximum reaching the trigger at all, and 390 of those are Class A. *This is why every rank-passing crossing in the survey is Class A*, and why the choice of reference class largely dissolves once the trigger condition is imposed. An expectation conditioned on the rank alone predicts a different quantity — the number of windows in which a no-signal position would rank first — and is a pipeline diagnostic rather than a statement about candidates.

A second conditioning is informative and is reported beside it, on the same control ensemble: on trigger, localisation and attribution without the rank clause, 36.5 Class A windows are expected to trigger by chance, the visibility fit localises 0.986 of events at $T_{\star} \approx 5$ (0.977 through all three clauses, so the factor is the same either way), and the mask occupies 4 per cent of the searched band, giving *34.5 expected unattributed Class A crossings against 35 observed*. The rank clause is a one-in-(512+1) filter, so the two numbers are not alternative readings of one quantity: each is the expectation for the chain named beside it, and on either chain the observed population is accounted for.

The same quantity admits a third treatment, quoted for comparison and not as a further expectation. Resampling execution blocks with replacement and additionally applying the measured non-exchangeability of the stellar position as a factor $\times 1.4$ gives 1.3 expected over the 1614 windows with stored ensembles, 95 per cent interval 0–4. The difference is the tail factor and nothing else, and that factor is itself known only to 1.2–1.9; the probability of the observed count or more under it is 0.4 to one significant figure, and no inference is drawn from either. The count cannot be dismissed as an artefact of the screen's position-dependent rank bias: that bias places the star *low* in its control distribution, and a bias in that direction suppresses star-first outcomes rather than manufacturing them.

One trade is not available to us, and it should be stated plainly. An excess of this size needs only a small un-

modelled tail in the control ensemble, and Appendix B shows that one exists. But the same noise model sets the injection recovery fractions, so a tail large enough to remove the excess would make the quoted completeness optimistic by roughly the same factor. The two cannot be traded independently, and neither correction has been applied.

The control ensemble: its calibration, and its resolution

The screen's geometry is set in §4.1; its calibration and its resolution are the subject here. Under that geometry the uniform distribution exchangeability implies is not rejected in either array stratum — 597 12m windows and 1048 ACA windows, the latter with median rank 0.478 against the expected 0.5 — over the 1645 of 1651 windows that retain a control vector, the remaining 6 having none. The ACA windows are screened on the geometry measured from each observation's own antenna table and baselines rather than on a 12 m beam, and the array configuration does not enter the sensitivity. Two cautions keep this from being a proof of exchangeability. At these sample sizes the test resolves only $D \gtrsim 0.042$, so a smaller value is consistent with uniform rather than demonstrably uniform; and the diagnostic most directly sensitive to contamination of the inner controls, the excess of the innermost control bin over the outer annulus, is $+0.038$ with a bootstrap interval of -0.214 to $+0.282$, which spans zero and therefore neither confirms nor excludes a residual effect.

Neither form is a significance test, and the reason is structural. Ranking the star against 512 control positions drawn from the same field is a powerful way to decide which windows deserve examination, but $1/(512 + 1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble — the smallest rank it can express — and not the probability that noise produced the window. Against a survey-wide Bonferroni scale

$$\alpha_{\text{survey}} = \frac{0.05}{1651} = 3.0 \times 10^{-5}, \quad \frac{1.95 \times 10^{-3}}{\alpha_{\text{survey}}} = 64, \quad (\text{D1})$$

that floor is 64 times too coarse, and correlation widens it: neighbouring controls are correlated below the synthesised-beam scale, so a window's annulus carries only $N_{\text{eff}} \simeq 30$ –43 independent spatial trials and the ensemble resolves no better than 0.023–0.032. Drawing more controls does not rescue it, the primary beam bounding how many independent positions a single pointing contains. **The floor of Eq. D1 is the design's central limitation: no single window in this survey can be significant on the rank alone, which is why a technosignature claim here requires independent recurrence or independent astrophysical confirmation.** What a visibility-domain statistic buys instead is measurable on this survey's own data: in the image plane every rank-passing window sits at rank 1 of $512 + 1$ with no dynamic range left to separate them, whereas the visibility fit separates the same windows by 5.2σ and its strongest reaches 9.5σ .

The trials budget

A window holds 177–6 869 740 channel $\times$ drift cells (median 468), the survey 1.32×10^8 in total, and the cells

are not independent trials: Hanning smoothing correlates neighbouring channels at $\rho_1 = 2/3$ and $\rho_2 = 1/6$ and the drift grid is oversampled two to one, giving a combined over-count of 5.3. The control ensembles calibrate that budget without a Gaussian assumption: the same arithmetic predicts 476 of the 1651 windows carrying a control above T_{trig} and 455 do, agreeing to 4 per cent, with per-window control maxima tracking the iid-Gaussian prediction at a median ratio 0.995 (5th–95th percentile 0.91–1.11). Correcting for the over-count leaves the observed control maxima +6 and +8 per cent above prediction in the fine and coarse classes independently. Non-Gaussianity of that size is why the operative statement is the empirical rank and not the Gaussian budget.

The residual on-star excess is not zero, and it is reported as an open systematic rather than explained away. Correlation cannot move the expected number above the trigger, only its clumping, so the excess over an iid-Gaussian budget decomposes into cell multiplicity within a crossing window — a factor 1.4, which is what an over-count of 2.67 on the frequency axis predicts, one real threshold excursion being counted about three times — and the count of crossing windows, a factor 1.3 whose one-sided Poisson probability is 0.04. The residual runs one way and carries the same sign at the controls. The crossing list is dominated by CO(2→1) at 230.5 GHz toward disc hosts, and a residual excess of chance crossings at the tens-of-per-cent level cannot be excluded.

One stratum departs from uniformity and it is the one that matters. The fine-channel windows fail the $U(0,1)$ rank test at $p = 0.016$ ($n = 402$), because 12 of them contain an on-star peak above their own control maximum against 0.78 expected; the coarse stratum, three times larger and predicting 2.43 such windows, contains 0. A 15.6–31.25 MHz channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it, and the same reasoning says a genuine unresolved carrier would appear in Class A first. The departure is nonetheless genuine and survives removal of every fine window carrying a stellar crossing, so fine-window rank probabilities are optimistic for the class as a whole and not only where a crossing sits.

What is left is bounded, and the bounds are null. Ranking each control against the other 512-1 by the rule used for the star gives 220 672 pseudo-star ranks, uniform to the resolution the ensemble can express; the two-sample comparison of the real stellar ranks against the pooled pseudo-ranks returns $p = 0.38$; the Spearman correlation of control statistic against reconstructed radius has median 0.002 over the 1187 crossing-free windows, bounding any systematic radial suppression of the controls below 10 per cent in the mean; and the star’s own residual variance equals the controls’ to better than a per cent in the rank-passing windows. A local scramble test, which destroys anything coherent along a drift track while preserving each integration’s own spectral autocorrelation, carries 4 windows past the rank floor and changes no disposition. It fails outright on HD 48370 Band 6 — the field carries extended emission and imaging residuals that are coherent across integrations and that the scramble destroys by design — so no scrambled-noise probability is quoted there. Per-control noise records and gain are absent from the retained products, so those stratifications

are bounded only through these rank distributions.

RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has, and the coverage it does not have, are both set out here.

Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one, and the searched range is not above all allocated services: ITU Radio Regulations Article 5 carries active allocations throughout 90–275 GHz, and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard in this band is the 94.0–94.1 GHz Earth exploration-satellite (active) allocation used by spaceborne cloud radar — CloudSat and, since 2024, EarthCARE — which is a documented contaminant of ALMA Band 3. 1 of the 1651 searched windows overlaps that band, and it carries 0 threshold crossings and 0 rank-passing events, so no result here rests on it and no overpass timing needs to be checked. What does favour this band is that terrestrial transmitters are sparse compared with centimetre wavelengths, that the array sits at 5000 m on a site chosen partly for radio quietness, and, decisively, that an interferometer is not a total-power detector: a signal that is not in the far field at the phase centre does not fringe-track, but acquires a rapidly varying geometric phase across the array and is suppressed in the correlation rather than imaged.

What screening exists, stated with its coverage, because the coverage is uneven. Two screens are computed for every searched window from the released catalogue and are reproducible from it: the 94.0–94.1 GHz allocation test above, and the sky-frequency and intermediate-frequency tests below. The 512-position rank screen asks whether an excess is at the star or everywhere in the field, which is an interference test as well as a prioritisation, and it covers every window. Recurrence at the same tuning in an independent epoch covers the crossings for which a covering repeat block exists. The visibility fit asks whether an excess has the phase signature of a source at the phase centre, which a near-field emitter would not have, but it is a consistency check rather than a filter. Finally, the correlator flagging is the observatory’s own, replayed verbatim from each block’s delivered calibration script; it is not recorded per window in the release and cannot be reconstructed from it. *No per-channel interference flag of our own exists, and none can be recovered.*

Four measurements support the reasoning, and each is a test the population could have failed.

First, fixed sky frequency. Interference is a property of the observatory and not of the star, so a terrestrial carrier should appear at the same sky frequency in every window that covers it. Of the 2 unattributed events, 2 have a released crossing frequency; 32 windows toward 8–24 other stars in 74 other blocks cover those frequencies, and 0 of them carries a crossing in the same channel.

Second, residual clustering in sky frequency. A scale-aware scan statistic — the maximum count in any band of width 1 to 300 MHz, calibrated against a null in which each window’s peak is drawn on its own ALMA channel grid, since two windows sharing a standard tuning share an identical grid — is flat in cluster width and rejects 100.0 per cent of the time at twelve injected events. Ap-

plied to the 51 crossings with a released frequency it *does* find clustering, at $p = 0.0040$ (12 observed against 7.38 expected), and the cluster is the survey’s own molecular emission: its maximal band, 230.423–230.723 GHz, holds 12 crossings of which 9 lie within 50 km s^{-1} of CO(2→1), and removing the line-attributed crossings removes the clustering ($p = 0.111$ for the 40 unattributed). Equivalently, and with no choice of band or width at all, 11 of the 51 crossings lie inside the survey’s own $\pm 50 \text{ km s}^{-1}$ tube against 4.7 expected under the matched grid null, $p = 0.0007$. *The crossing population is significantly concentrated on the masked transitions, and that concentration is the attribution rather than a residue left over after it.*

Third, fixed intermediate frequency. An artefact of the signal chain would sit at a fixed intermediate frequency rather than a fixed sky frequency, and the crossings do not: their fractional positions within their own windows have median 0.50 and are uniform over 1397 crossings.

Fourth, a cross-target occupancy screen over every searched window and not only the crossings. A terrestrial carrier should recur at the same sky frequency toward *unrelated* stars. Over all 3027 searched windows toward 133 stars, with each window’s peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($p = 0.15$); among the 109 windows whose peak exceeds 5σ it is 4 against 5.2 ± 1.9 , i.e. *below* the null, and with the molecular-line mask applied 1 against 2.1 ± 1.3 . There is no excess to explain. The screen is keyed on the

star rather than on the product directory, which matters: a star’s own repeat observations are not different targets, and every persistent feature — a real line most of all — would otherwise enter as a cross-target coincidence. 35 of those 109 windows sit on a masked transition (CO(2→1) $\times 17$, CO(1→0) $\times 10$, CO(3→2) $\times 4$, $^{13}\text{CO}(2\rightarrow 1) \times 3$, SiO(5→4) $\times 1$), which is the point: a CO line falls at almost the same topocentric frequency toward every nearby star, so cross-star frequency coincidence is exactly what real line emission produces, and the screen cannot distinguish it from interference unless the mask is applied first.

One crossing is rejected as instrumental on these grounds: BD+05 1668, which appears in the same coarse channel toward unrelated stars and whose visibility fit places it far off the phase centre.

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. Accordingly we claim only that *no identified interference mechanism accounts for these events*: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and an archival dataset without a simultaneous off-source strategy cannot do better. What the four tests establish is that the crossing population as a whole carries none of the signatures interference leaves — fixed sky frequency, fixed intermediate frequency, or presence in unrelated targets — which is a statement about the population and not a clearance of any individual crossing.

REFERENCES

- Astudillo-Defru N., et al., 2017, A&A, 602, A88
 Astropy Collab., 2022, ApJ, 935, 167
 Backus P. R., et al., 2002, BAAS, 34, 1173
 Blomme R., et al., 2023, A&A, 674, A7
 CASA Team, 2022, PASP, 134, 114501
 Burton K., MacGregor M. A., Osten R. A., 2022, ApJL, 939, L6
 Burton K., 2024, PhD thesis, Univ. Colorado Boulder, ch. 4
 Cataldi G., et al., 2023, ApJ, 951, 111
 Choza C., et al., 2024, AJ, 168, 254
 Cocconi G., Morrison P., 1959, Nature, 184, 844
 Czech D. J., et al., 2021, PASP, 133, 064502
 Dent W. R. F., et al., 2014, Science, 343, 1490
 Drake F. D., 1961, Physics Today, 14(4), 40
 Drake F. D., 1974, in Sagan C., ed., CETI. MIT Press
 Enriquez J. E., et al., 2017, ApJ, 849, 104
 Gaia Collab., Vallenari A., et al., 2023, A&A, 674, A1
 Garrett M. A., et al., 2026, MNRAS, in press (arXiv:2605.10212)
 Gentile Fusillo N. P., et al., 2021, MNRAS, 508, 3877
 Ginsburg A., et al., 2019, AJ, 157, 98
 Harris C. R., et al., 2020, Nature, 585, 357
 Huang B.-L., et al., 2026, ApJ, 1006, 9
 Jacobson-Bell K., et al., 2025, AJ, 169, 233
 Johnson O. A., et al., 2023, AJ, 166, 193
 Li J.-K., et al., 2026, arXiv:2603.19023 (preprint)
 Ma P. X., et al., 2023, NatAs, 7, 492
 MacGregor M. A., et al., 2020, ApJ, 891, 80
 Manunza L., et al., 2025, AcAau, 233, 155
 Margot J.-L., et al., 2023, AJ, 166, 206
 Mason L. A., Garrett M. A., Wandia K., Siemion A. P. V., 2025, MNRAS, 536, 2127 (arXiv:2411.19827)
 Matrà L., et al., 2017, MNRAS, 464, 1415
 Morrison I. S., et al., 2023, PASA, 40, e038
 Müller H. S. P., et al., 2005, JMoSt, 742, 215
 Ortiz Ceballos K., et al., 2026, arXiv:2609.16720 (preprint)
 Pickett H. M., et al., 1998, JQSRT, 60, 883
 Poznanski D., et al., 2025, AJ, in press (arXiv:2505.03927)
 Price D. C., et al., 2020, AJ, 159, 86
 Sheikh S. Z., et al., 2025, AJ, 169, 118
 Sheikh S. Z., 2020, IJAsB, 19, 237
 Sheikh S. Z., et al., 2021, Nature Astronomy, 5, 1153
 SpaceX, 2024, FCC IBFS SAT-MOD-20230207-00021
 Tarter J., 2001, ARA&A, 39, 511
 Tremblay C. D., et al., 2024, PASA, 41, e004
 Vidal C., et al., 2026, arXiv:2605.21093 (preprint)
 Virtanen P., et al., 2020, NatMe, 17, 261
 White G. J., et al., 2026, MNRAS, submitted
 Wright J. T., et al., 2018, AJ, 156, 260
 Zhang Z.-S., et al., 2020, ApJ, 891, 174